MODULE 4 LAB REPORT

Perimeter Network Security

Perimeter Network Security with pfSense, Suricata IDS & Attack Simulation

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1. Introduction

Module 4 extends the cybersecurity lab built in Modules 2 and 3 by adding a complete perimeter network security layer. This module introduces pfSense as the network gateway and edge firewall, replacing the default gateway on all existing virtual machines.

The objectives of this module are:

- Deploy pfSense as the central edge firewall and router for the lab network
- Configure firewall rules to enforce traffic policies across the internal network
- Update Suricata IDS on IDS01 with custom detection rules for SQL injection, XSS, brute force, and LLMNR poisoning
- Run a full attack simulation from ATTACK01 to validate every security control
- Verify that Active Directory, GPOs, and VPN services remain functional after the gateway change

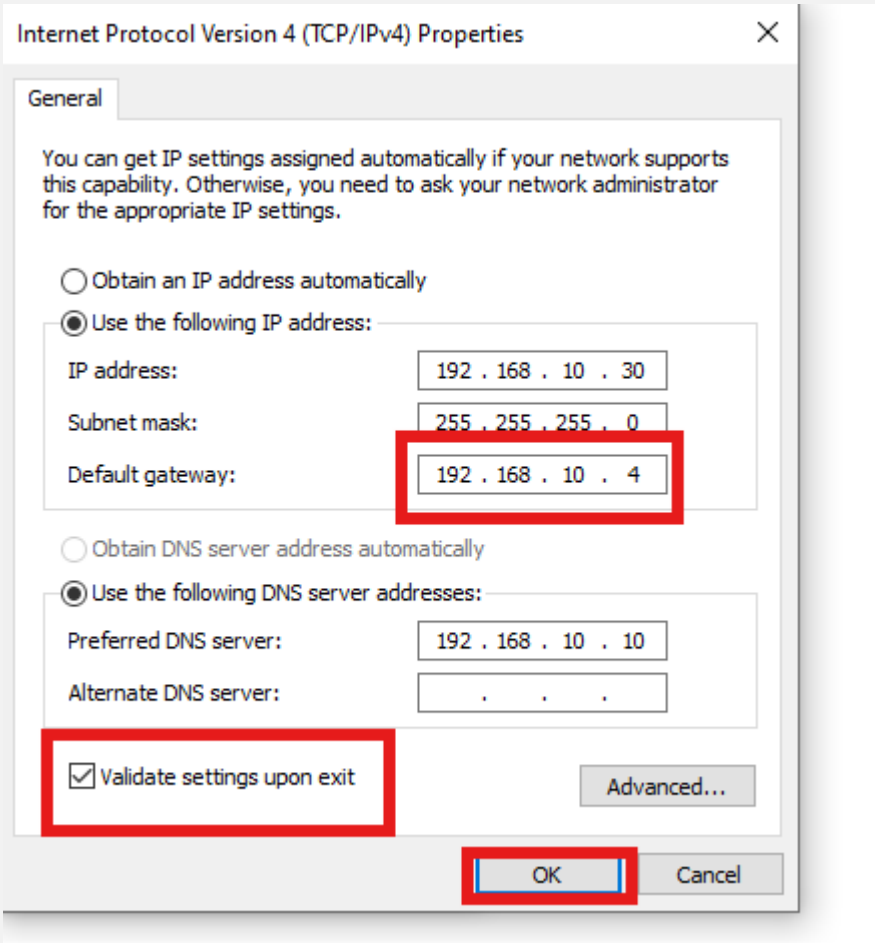
Together with Modules 2 and 3, this module demonstrates a practical implementation of defence in depth across administrative, technical, and perimeter security controls.

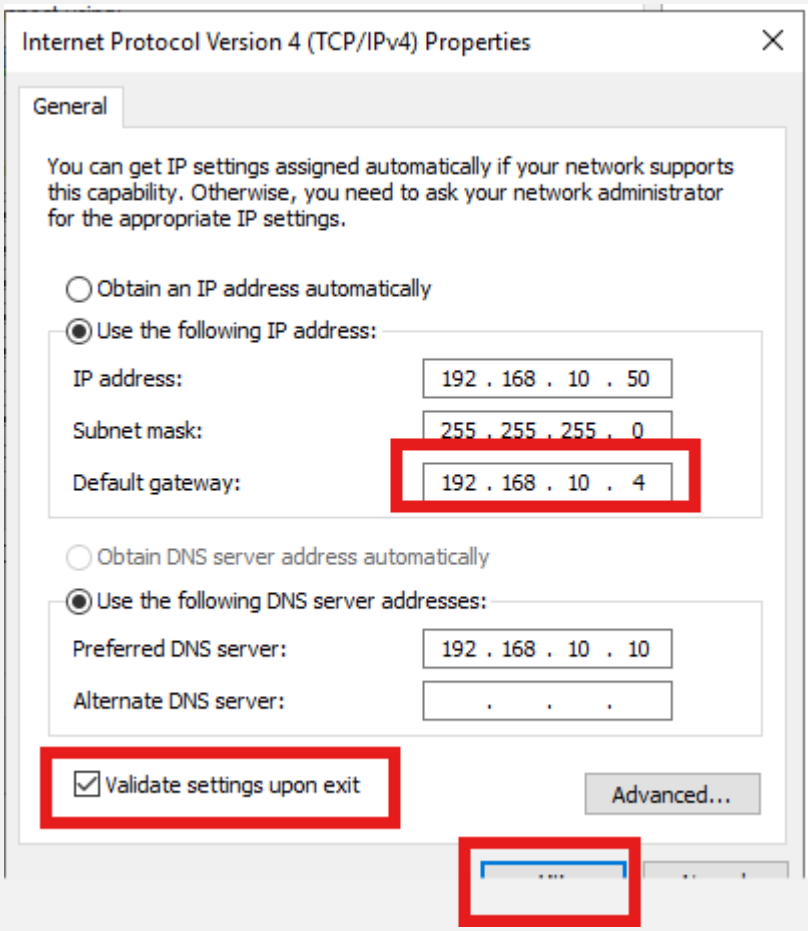
2. Lab Environment Overview

2.1 Virtual Machines

The following virtual machines form the complete lab environment for Module 4:

Machine	IP Address	OS / Platform	Role
pfSense	WAN: NAT | LAN: 192.168.10.4	pfSense (FreeBSD)	Edge Firewall & Router
DC01	192.168.10.10	Windows Server	Domain Controller (Active Directory)
ADMIN01	192.168.10.20	Windows	Admin Workstation / pfSense Web UI
CL01	192.168.10.30	Windows	Client Machine (Firewall Rule Testing)

Machine	IP Address	OS / Platform	Role
			
ATTACK01	192.168.10.40	Kali Linux	Attacker Machine (All Attacks)
VPN01	192.168.10.50	Windows	SoftEther VPN Server

Machine	IP Address	OS / Platform	Role
			
IDS01	192.168.10.60	Kali Linux	Intrusion Detection System (Suricata)

2.2 Network Architecture

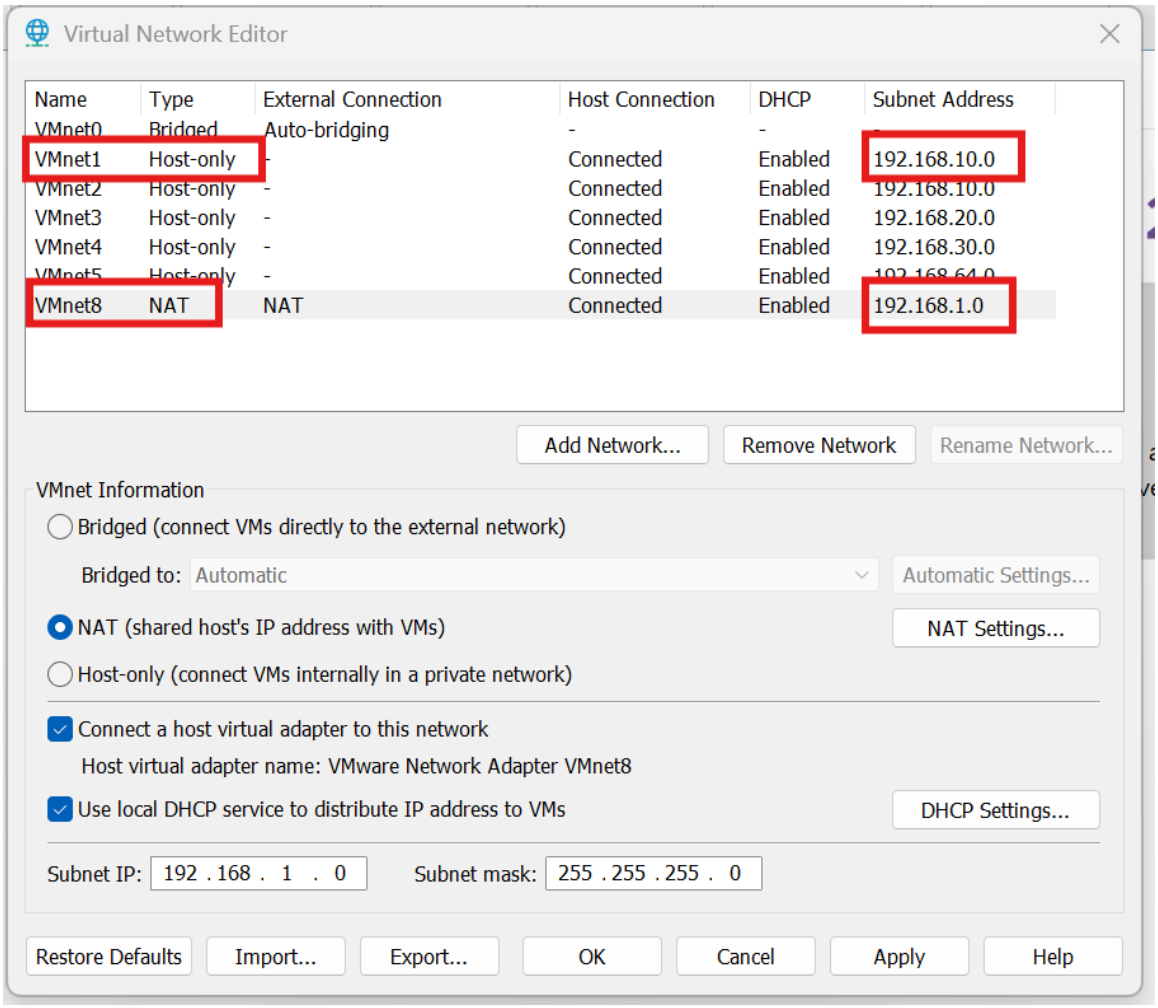
Network Segment	Subnet	Purpose
VMnet1 (Host-Only)	192.168.10.0/24	Internal LAN — all VMs communicate here
VMnet8 (NAT)	192.168.1.0/24	WAN — pfSense internet access
VPN Clients	192.168.30.0/24	SoftEther VPN client address pool

3. pfSense Configuration

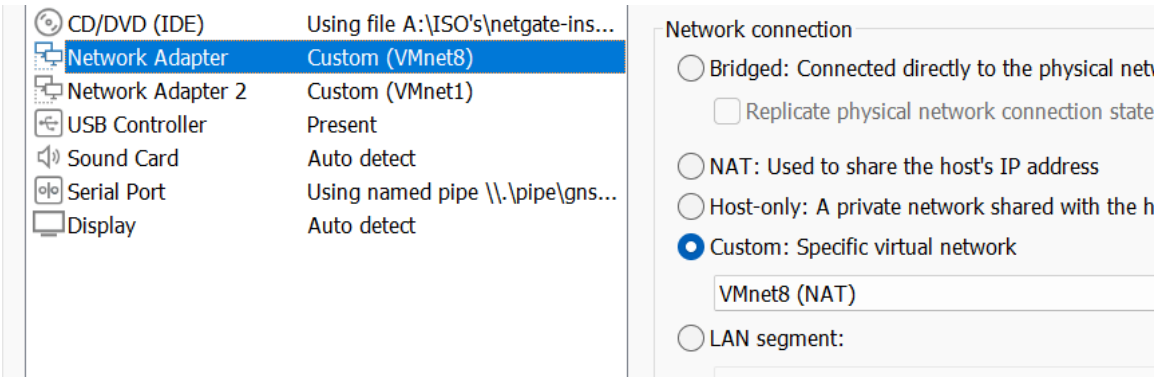
3.1 VMware Network Preparation

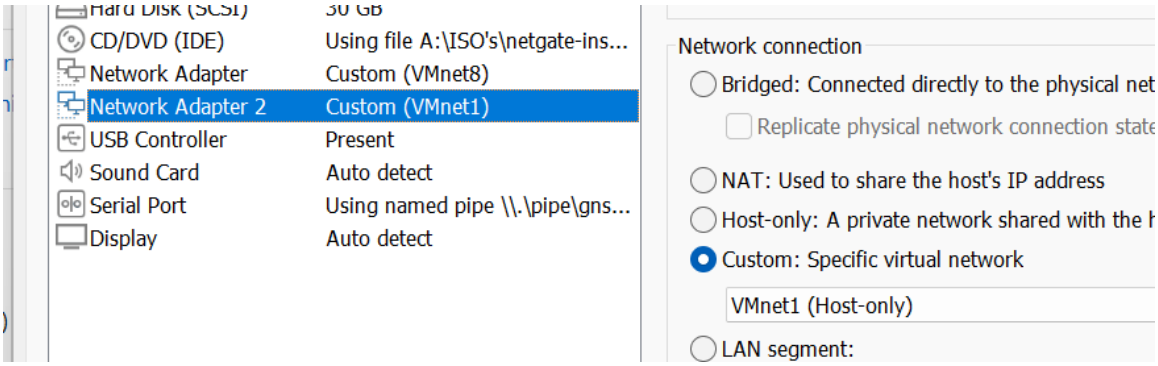
Before deploying pfSense, the VMware virtual network adapters were configured:

- VMnet1 set to Host-Only mode, subnet 192.168.10.0/24, DHCP disabled (static IPs assigned manually)
- VMnet8 left as NAT (192.168.1.0/24) to provide pfSense with WAN/internet access
- pfSense VM: Adapter 1 set to NAT (WAN), Adapter 2 set to Host-Only/VMnet1 (LAN)



- Network Adapter Setup





3.2 pfSense LAN IP Configuration

From the pfSense console (Option 2 — Set interface IP address):

```
Interface: LAN
IP Address: 192.168.10.4
Subnet Mask: /24 (255.255.255.0)
DHCP Range: 192.168.10.100 - 192.168.1
0.200
```

- Pfsense Lan setup

```
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on pfSense ***

 WAN (wan) -> em0 -> v4/DHCP4: 192.168.1.128/24
 LAN (lan) -> em1 -> v4: 192.168.10.4/24

 0) Logout / Disconnect SSH            9) pfTop
 1) Assign Interfaces                 10) Filter Logs
 2) Set interface(s) IP address       11) Restart GUI
 3) Reset admin account and password  12) PHP shell + pfSense tools
 4) Reset to factory defaults         13) Update from console
 5) Reboot system                     14) Enable Secure Shell (sshd)
 6) Halt system                       15) Restore recent configuration
 7) Ping host                         16) Restart PHP-FPM
 8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2
```

```
Configure IPv4 address LAN interface via DHCP? (y/n) n

Enter the new LAN IPv4 address.  Press <ENTER> for none:
> 192.168.10.4
```

```
      255.255.0.0    = 16
      255.0.0.0      = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via DHCP6? (y/n) n

Enter the new LAN IPv6 address.  Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) y
Enter the start address of the IPv4 client address range: 192.168.1.100
This IP address must be in the interface's subnet
Enter the start address of the IPv4 client address range: 192.168.1.100
This IP address must be in the interface's subnet
Enter the start address of the IPv4 client address range: 192.168.10.100
Enter the end address of the IPv4 client address range: 192.168.10.200
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n)
```

```
Do you want to revert to HTTP as the webConfigurator protocol? (y/n) n

Please wait while the changes are saved to LAN...
 Reloading filter...
 Reloading routing configuration...
 DHCPD...

The IPv4 LAN address has been set to 192.168.10.4/24
You can now access the webConfigurator by opening the following URL in your web
browser:
              https://192.168.10.4/

Press <ENTER> to continue.
```

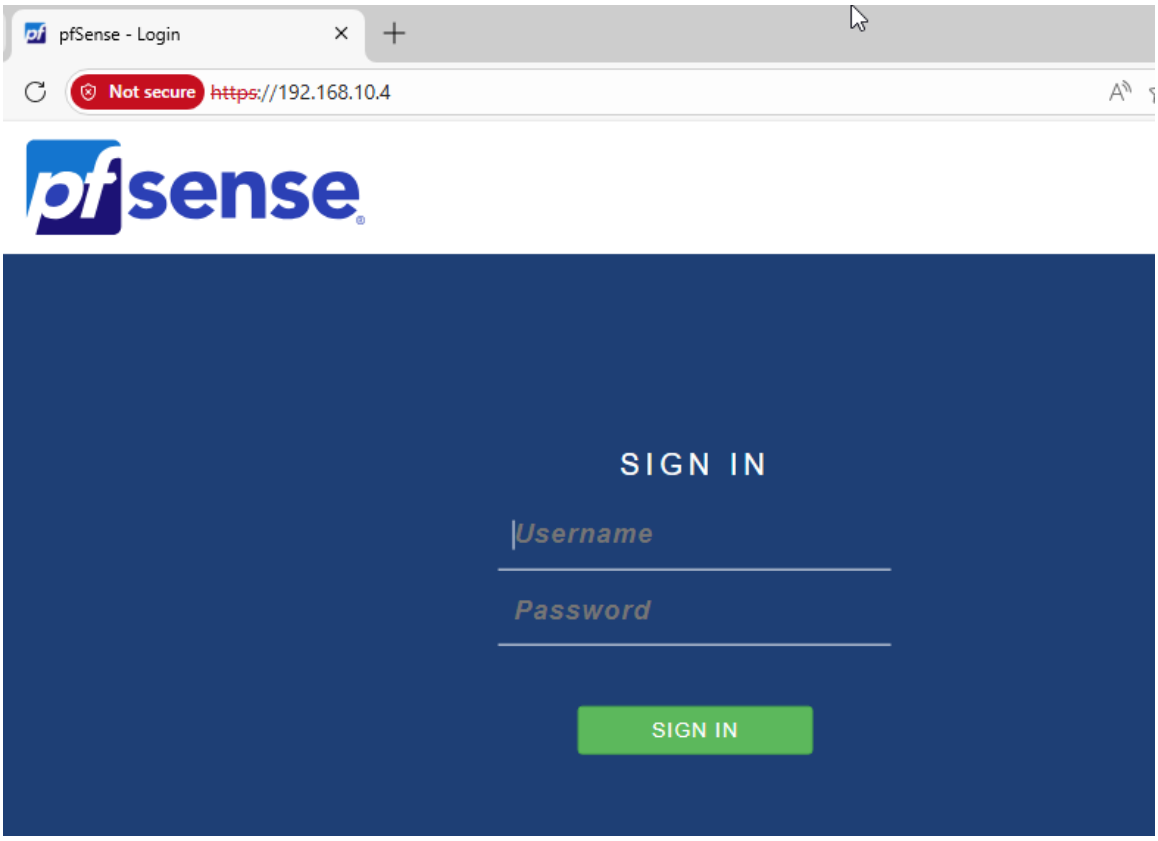
3.3 pfSense Web UI Setup Wizard

Accessed from ADMIN01 browser at https://192.168.10.4. Setup wizard configured with:

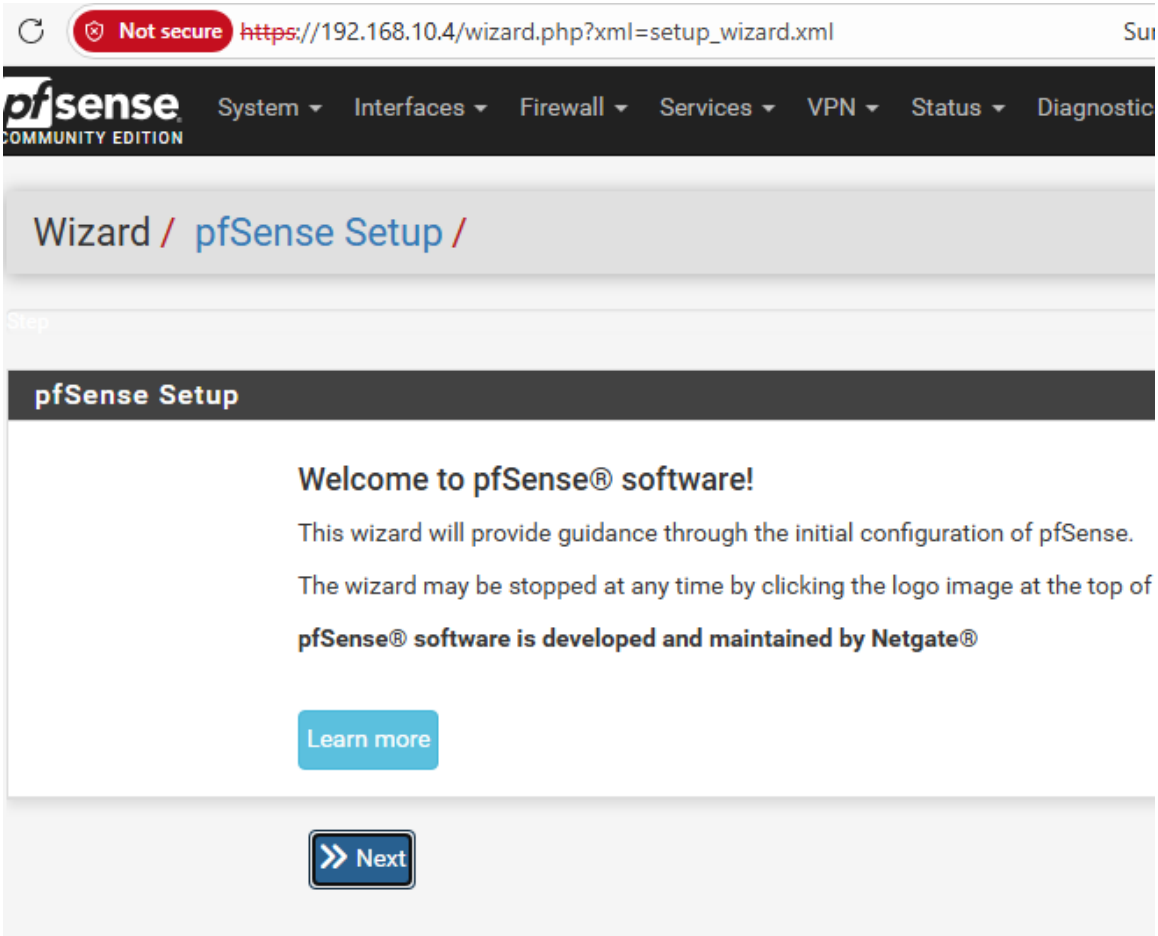
- Domain: corp.local (matching Modules 2 & 3 configuration)
- Primary DNS: 192.168.10.10 (DC01)
- Timezone: set to local timezone
- WAN: left as default NAT configuration
- LAN IP: confirmed as 192.168.10.4
- Admin password: changed from default pfsense to secure password

Result: pfSense dashboard confirmed WAN IP (from NAT) and LAN IP 192.168.10.4 active.

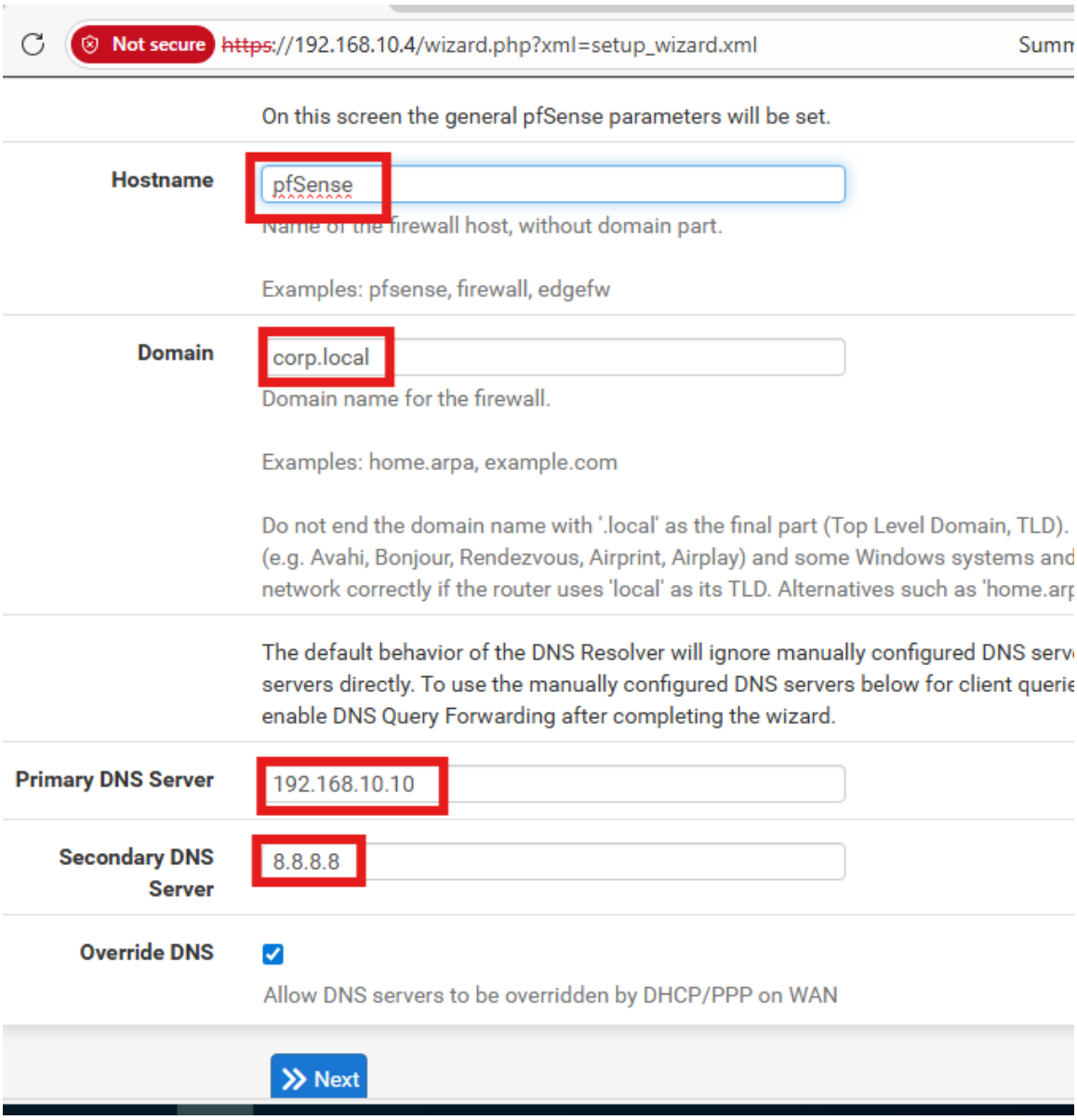
- Wizard Installation



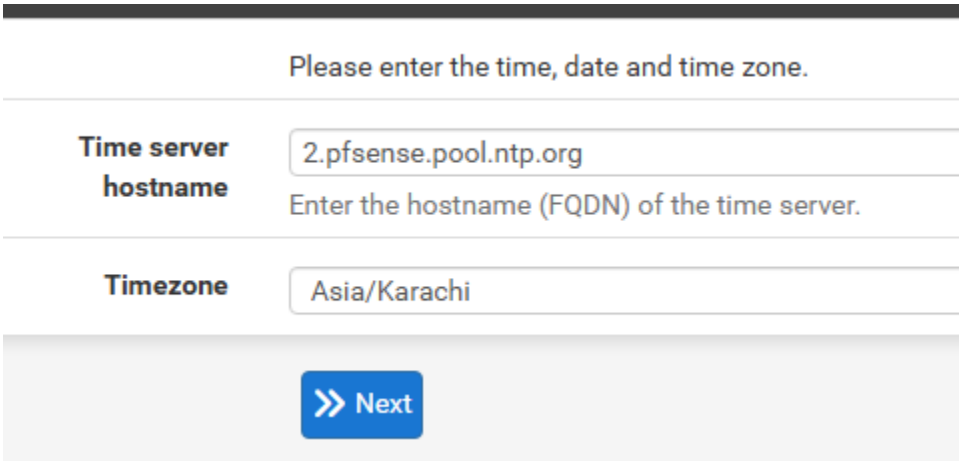
- Click on next to move forward with the installation



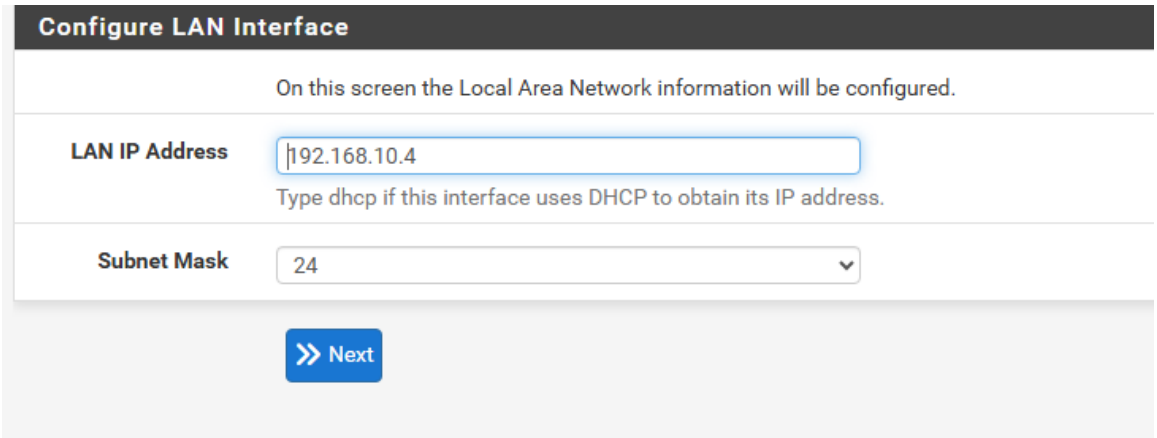
- Setup the Hostname, Domain and Primary DNS Server



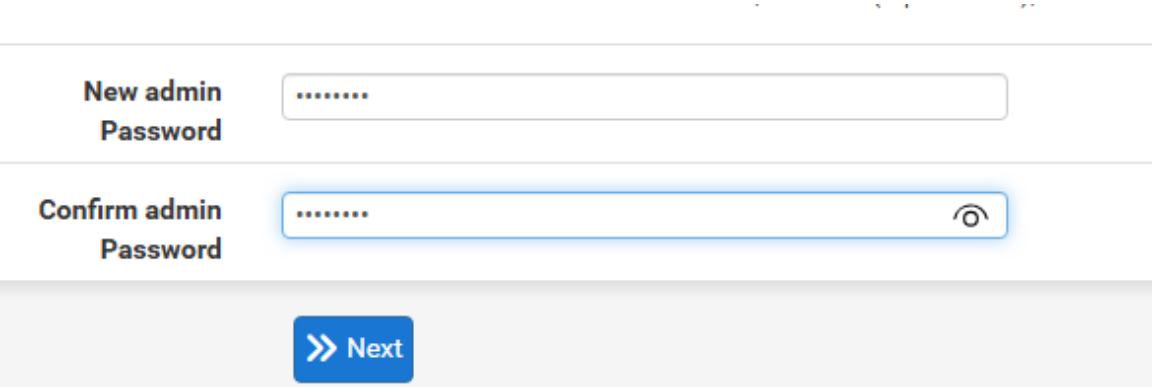
- Time zone setup



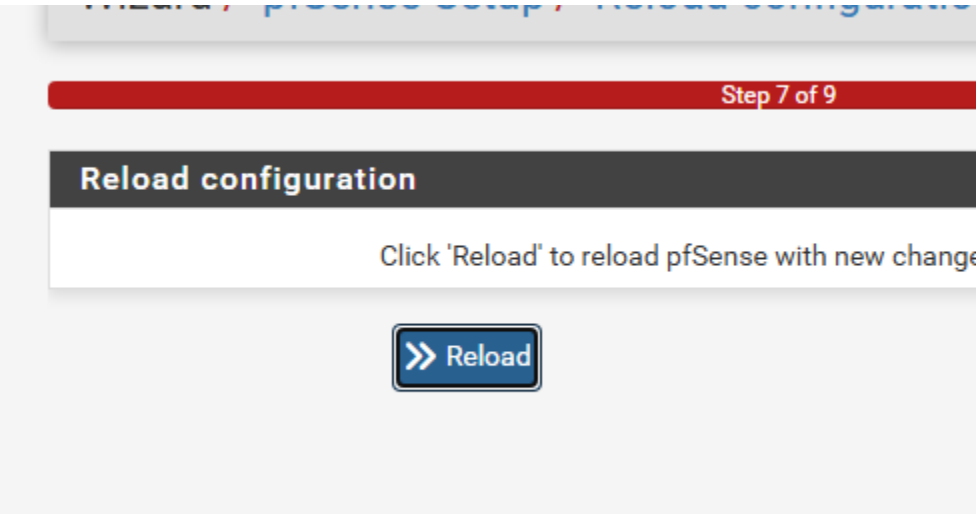
- Configuring LAN interface



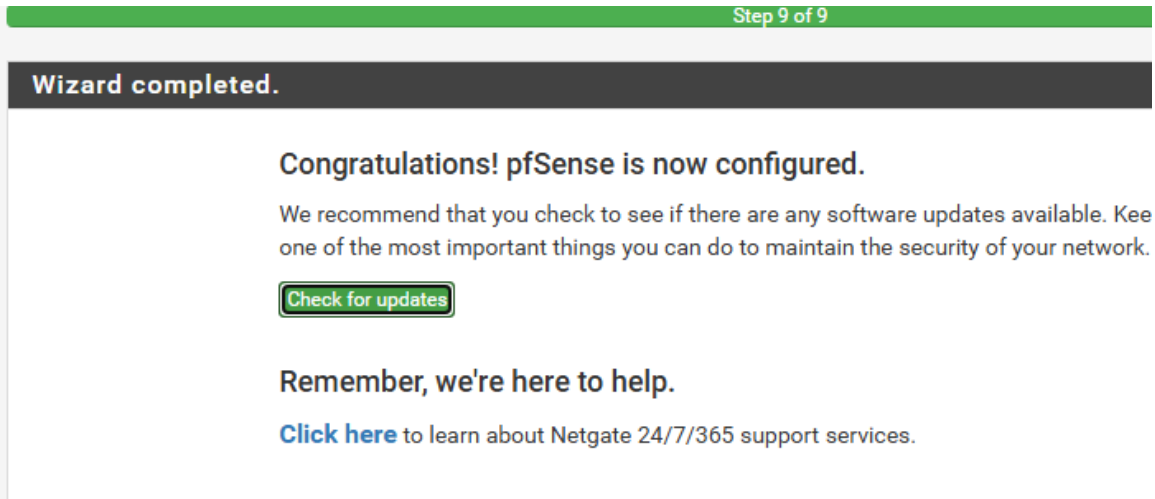
- New Password Setup



- Reload the configuration for the changes to take place.



- Wizard Setup completed

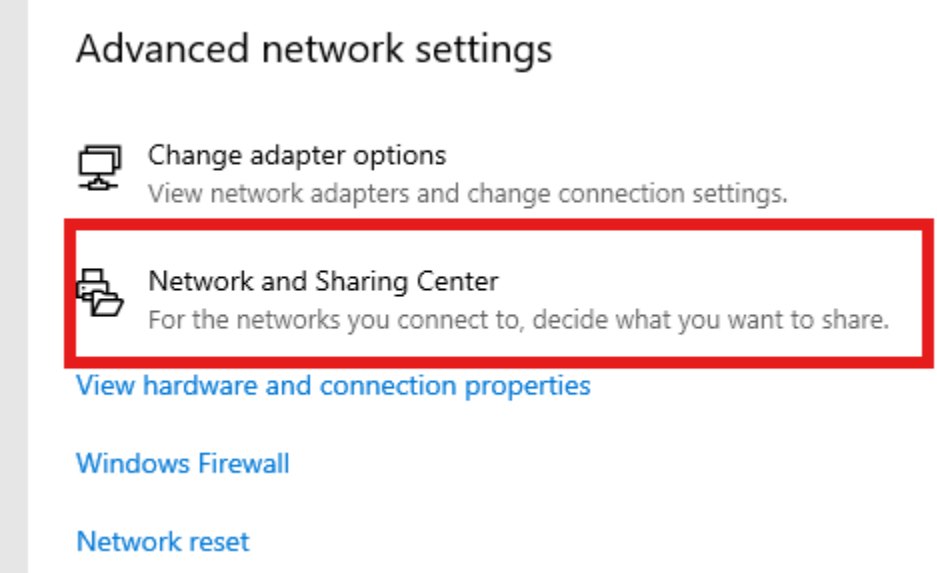


4. Gateway Update on All VMs

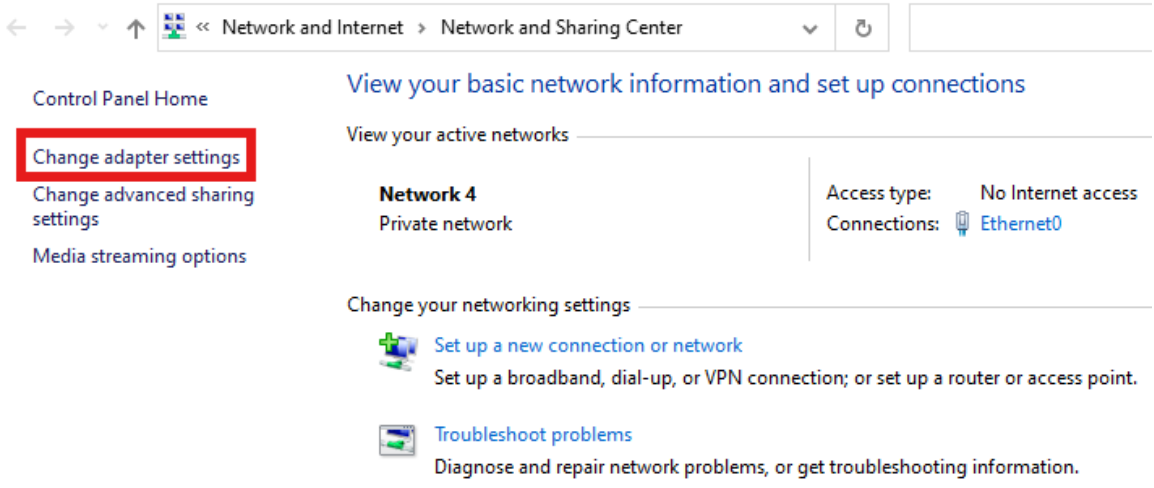
> **Critical Step: Every VM must point to 192.168.10.4 as its default gateway. Without this, traffic does not route through pfSense and firewall rules have no effect.**

4.1 Windows VMs (DC01, ADMIN01, CL01, VPN01)

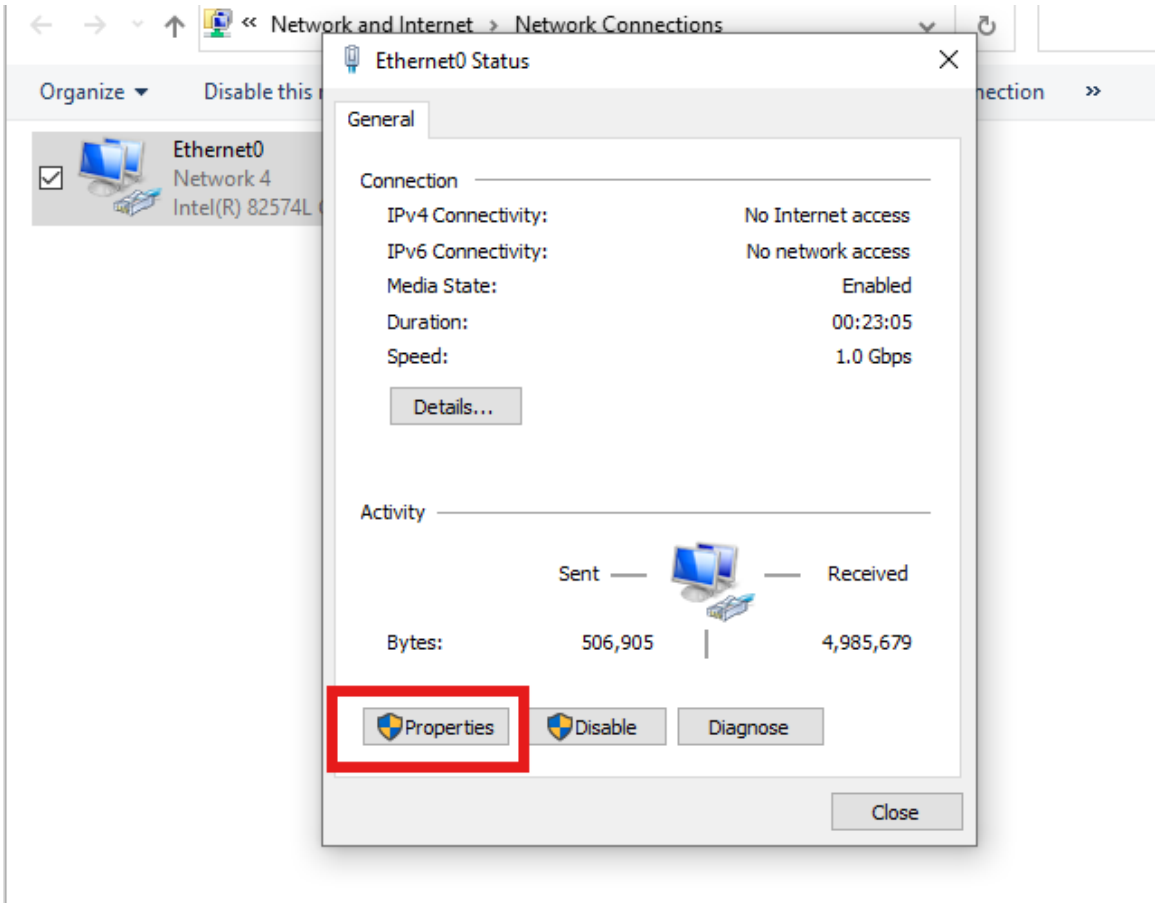
Steps performed on each Windows VM: Network and Sharing Center > Change adapter settings > Right-click Ethernet > Properties > IPv4 > Change Default gateway to 192.168.10.4.



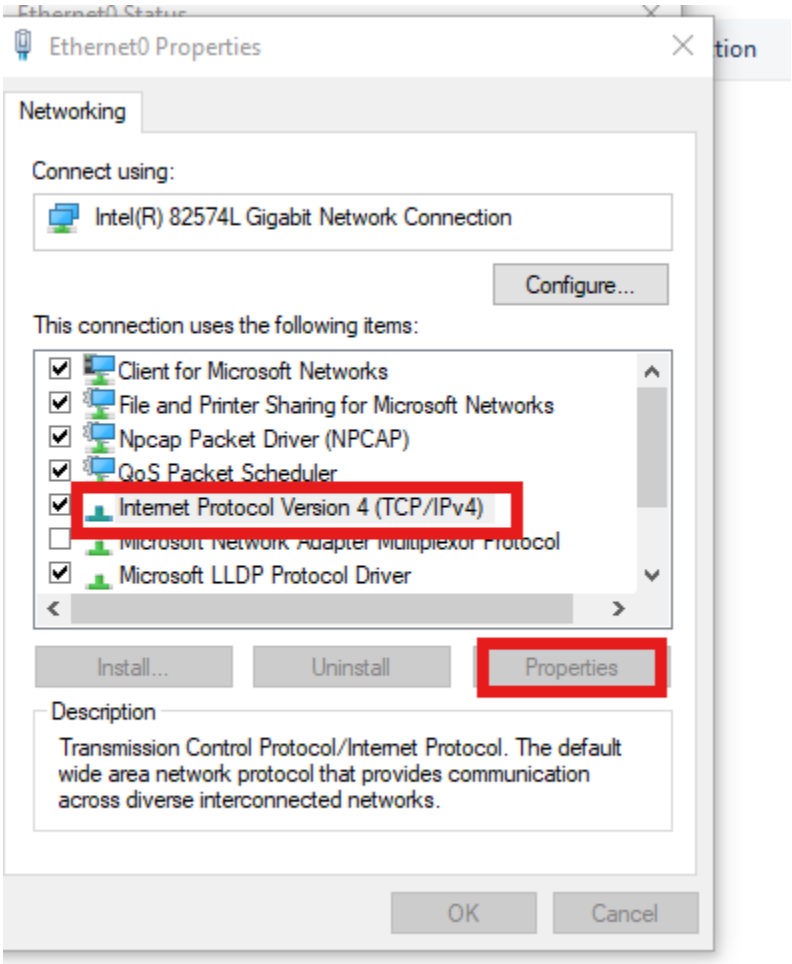
- Click on Change Adapter settings



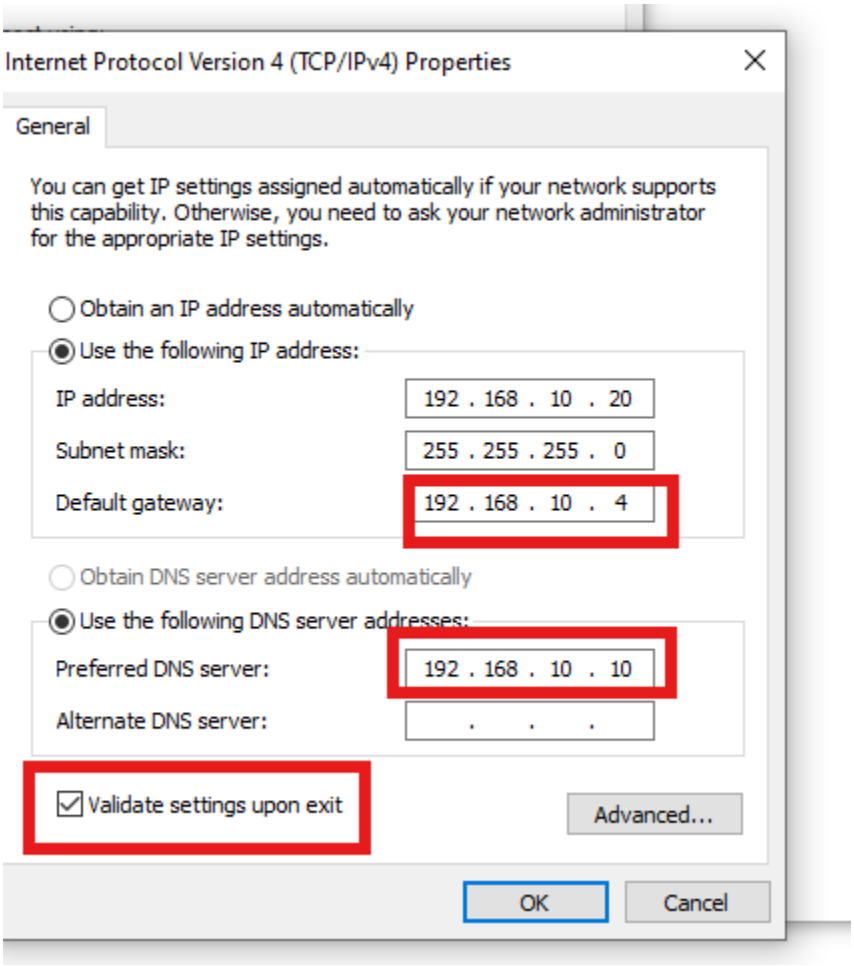
- Click on Properties



- Click on IPv4 Settings.

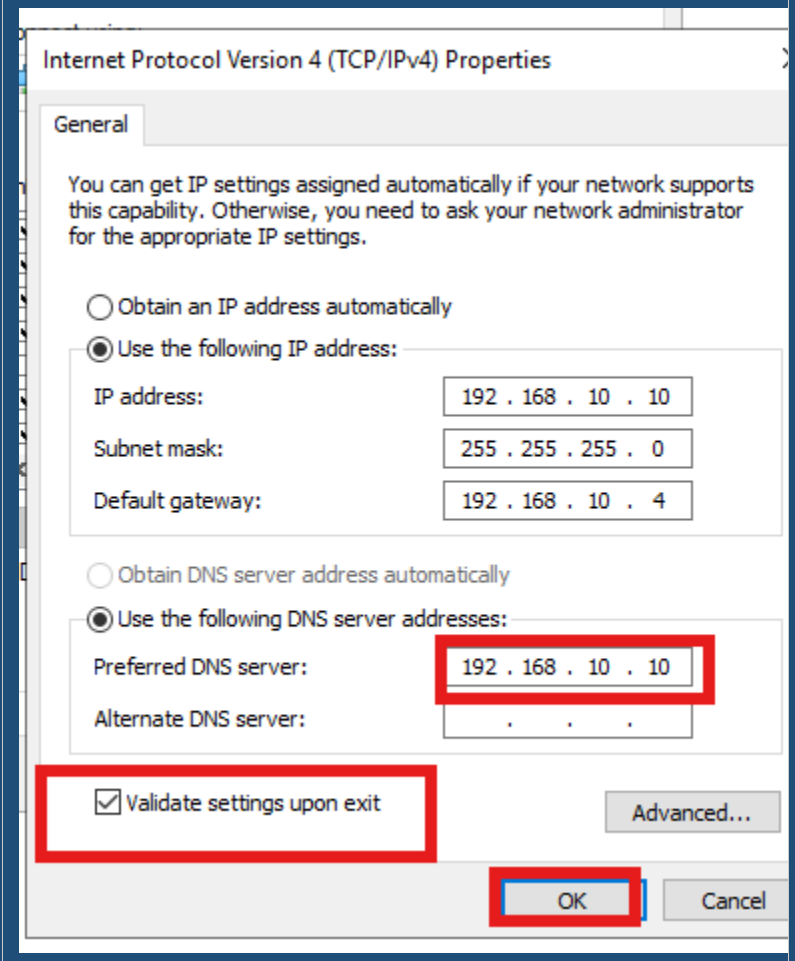


- Set Static IP Address and the preferred DNS server address



- Check the validate settings upon exit box for the changes to take place

Machine	Static IP	New Gateway	DNS
DC01	192.168.10.10	192.168.10.4	192.168.10.10
ADMIN01	192.168.10.20	192.168.10.4	192.168.10.10
CL01	192.168.10.30	192.168.10.4	192.168.10.10
VPN01	192.168.10.50	192.168.10.4	192.168.10.10



4.2 Kali Linux VMs (IDS01, ATTACK01)

Configured using nmcli on each Kali machine:

```
sudo nmcli con mod "Wired connection 1" \
  ipv4.addresses 192.168.10.60/24 \   # IDS01 (use .40 for ATTACK01)
  ipv4.gateway 192.168.10.4 \
  ipv4.dns 192.168.10.10 \
  ipv4.method manual
sudo nmcli con up "Wired connection 1"
```

- Set the kal Linux IP address

```
┌──(cyber㉿cyber)-[~]
└─$ sudo nmcli con mod "Wired connection 1"\
> ipv4.addresses 192.168.10.40/24\
> ipv4.gateway 192.168.10.4\
> ipv4.dns 192.168.10.10\
> ipv4.method manual
```

- Enter the commands using ncmli

```
┌──(cyber㉿cyber)-[~]
└─$ sudo nmcli con mod "eth0" \
ipv4.addresses 192.168.10.40/24 \
ipv4.gateway 192.168.10.4 \
ipv4.dns 192.168.10.10 \
ipv4.method manual

┌──(cyber㉿cyber)-[~]
└─$ sudo nmcli con up "eth0"
Connection successfully activated (D-Bus active path: /org/freedesktop/NetworkManager/ActiveConnection/3)

┌──(cyber㉿cyber)-[~]
└─$ ip route
default via 192.168.10.4 dev eth0 proto static metric 100
192.168.10.0/24 dev eth0 proto kernel scope link src 192.168.10.40 metric 100
```

4.3 Connectivity Verification

After updating all gateways, connectivity was verified by pinging pfSense from ADMIN01:

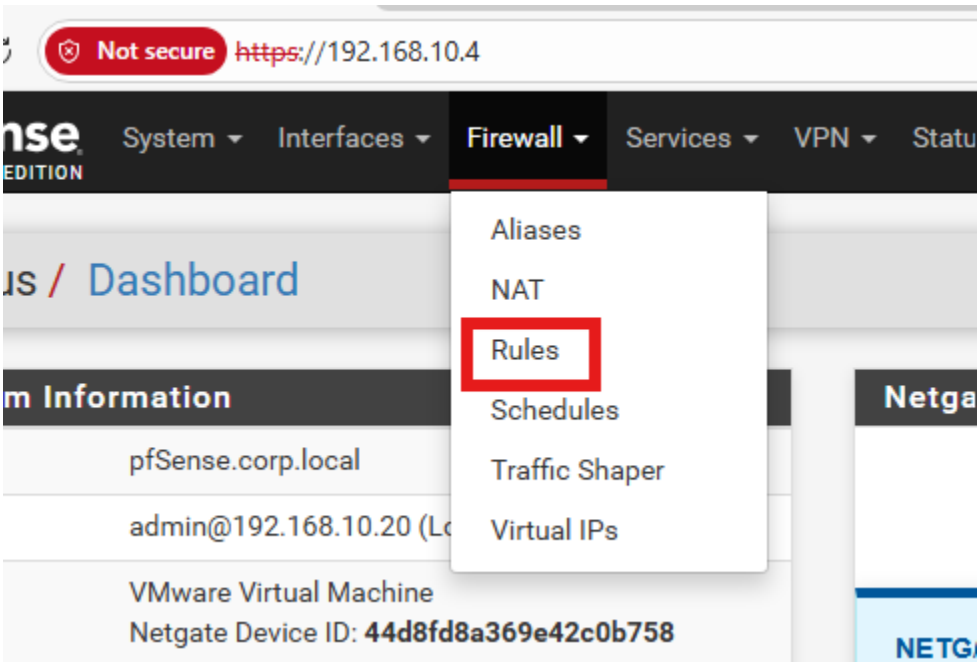
```
ping 192.168.10.4
```

Result: Replies received from pfSense on all machines. Gateway update confirmed working.

```
C:\Users\Administrator>ping 192.168.10.4

Pinging 192.168.10.4 with 32 bytes of data:
Reply from 192.168.10.4: bytes=32 time<1ms TTL=64
Reply from 192.168.10.4: bytes=32 time=1ms TTL=64
Reply from 192.168.10.4: bytes=32 time=1ms TTL=64
Reply from 192.168.10.4: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.10.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\Users\Administrator>
```

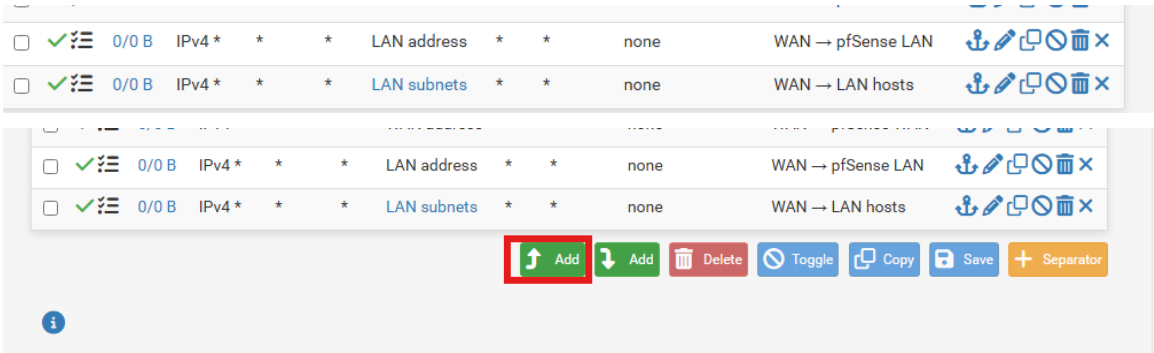
5. Firewall Rules Configuration

All firewall rules were configured in pfSense Web UI at https://192.168.10.4 under Firewall > Rules > LAN tab. pfSense processes rules top to bottom — block rules placed above allow rules.



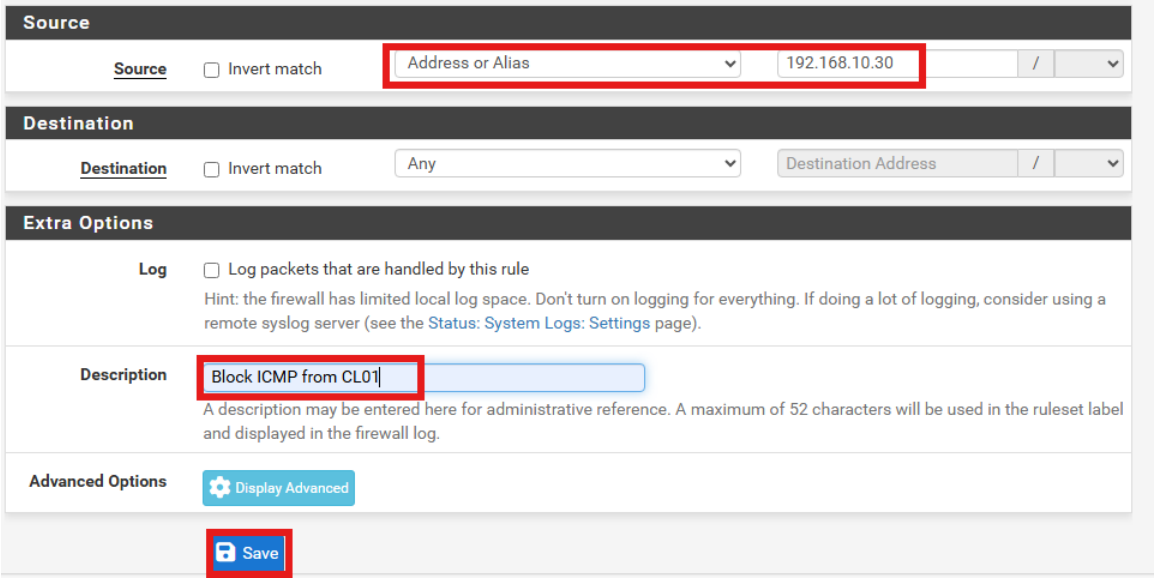
5.1 Rule 1 — Block ICMP from CL01

- Click on the add button showing the above direction arrow for the rule to take highest priority



Setting	Value
Action	Block
Protocol	ICMP — Any Type
Source	192.168.10.30 (CL01)

Setting	Value
Destination	Any
Description	Block-ICMP-From-CL01

Setting	Value
	
Logging	Enabled

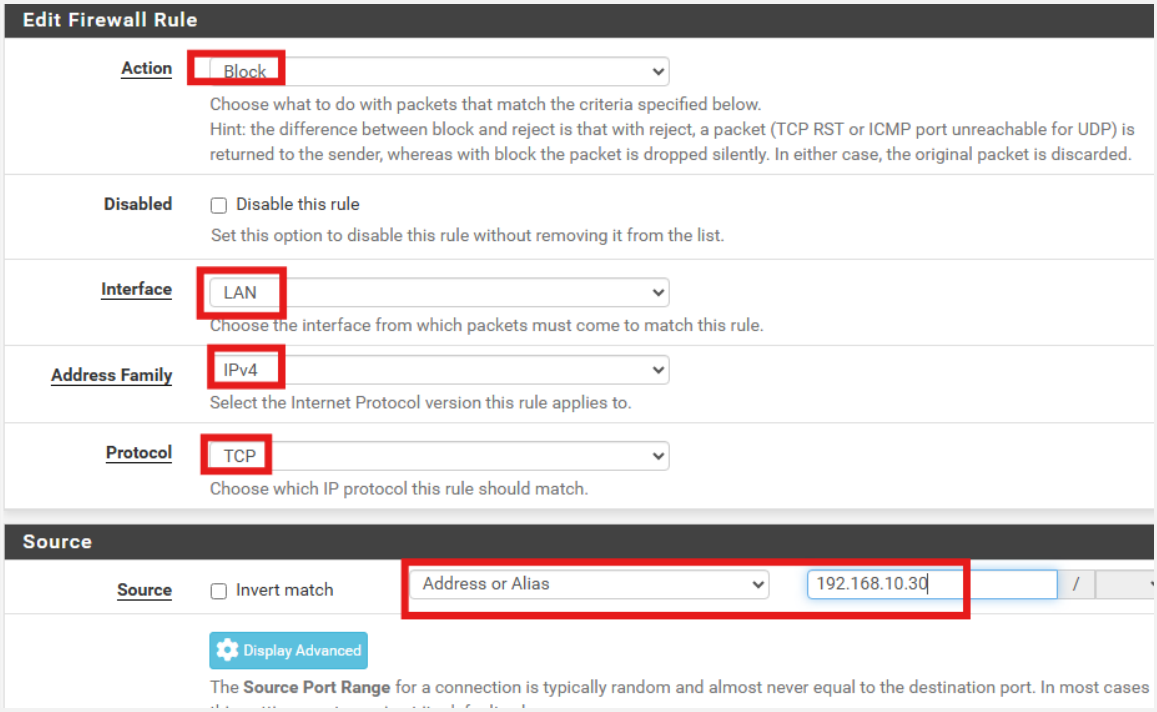
Test Result: ping from CL01 to 8.8.8.8 returned "Destination host unreachable" — rule working correctly. Replies came from 192.168.10.30 (local machine), confirming packet was blocked before leaving the network.

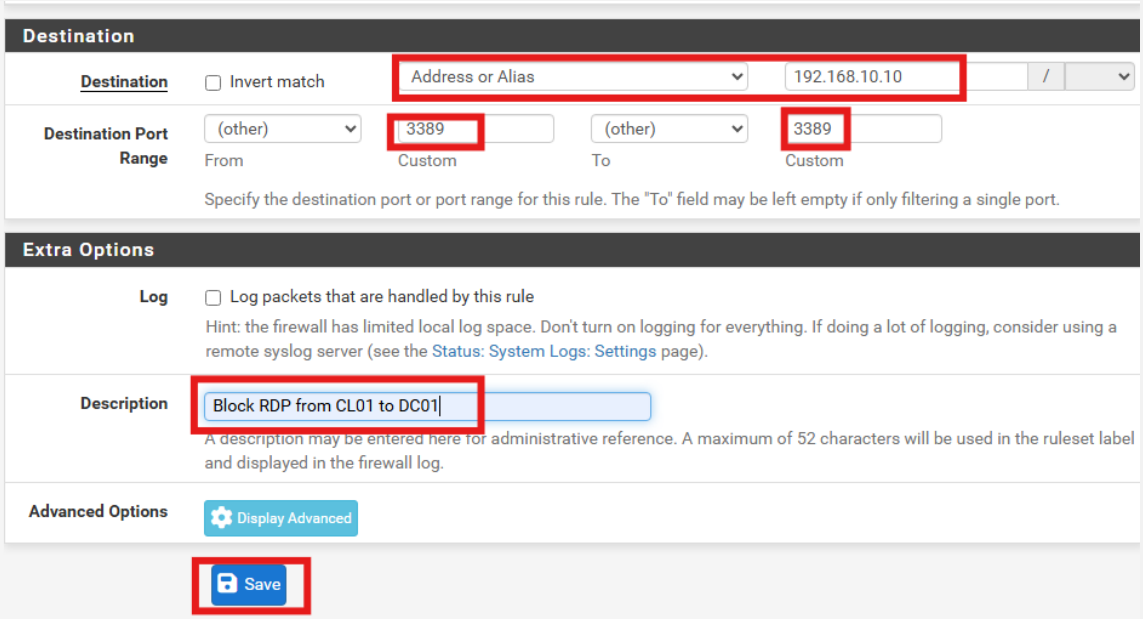
```
C:\Users\Administrator>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 192.168.10.30: Destination host unreachable.
Reply from 192.168.10.30: Destination host unreachable.
Reply from 192.168.10.30: Destination host unreachable.
Reply from 192.168.10.30: Destination host unreachable.

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

5.2 Rule 2 — Block RDP from CL01 to DC01

Setting	Value
Action	Block
Protocol	TCP
Source	192.168.10.30 (CL01)
Destination	192.168.10.10 (DC01)

Setting	Value
Destination Port	3389 (RDP)
Description	Block-RDP-CL01-to-DC01
Logging	Enabled

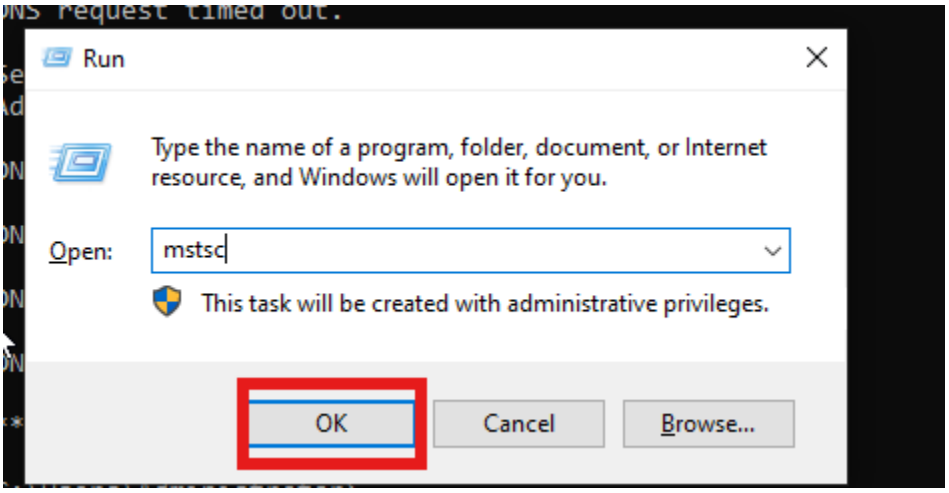




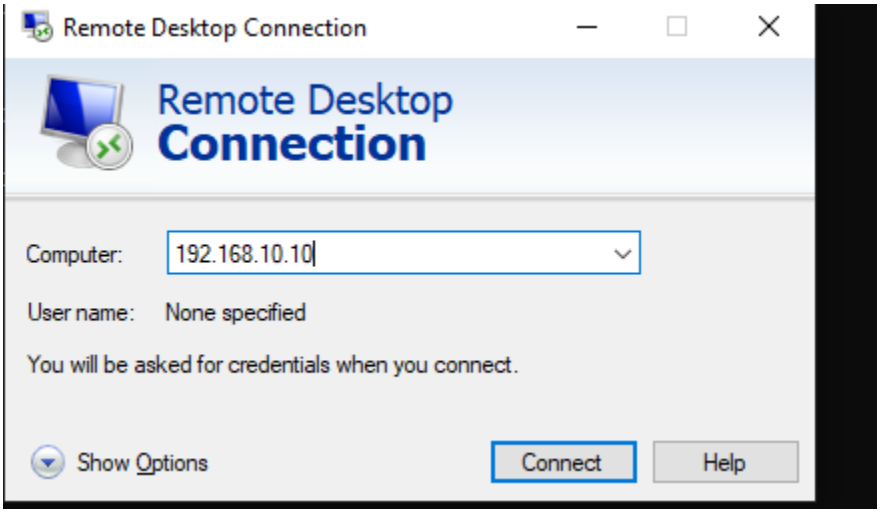
Test Result: RDP connection attempt from CL01 to DC01 (mstsc > 192.168.10.10) failed as expected.

```
C:\Users\Administrator>ping 192.168.10.10

Pinging 192.168.10.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.10.30: Destination host unreachable.
Request timed out.
Reply from 192.168.10.30: Destination host unreachable.

Ping statistics for 192.168.10.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
```

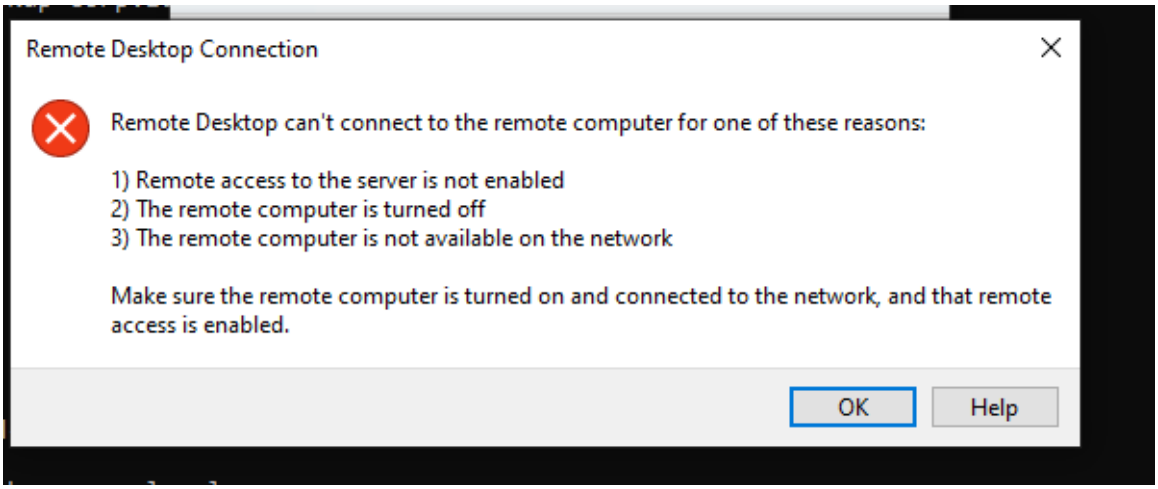
- Press Win + R and search mstsc and press Enter



- Set the Computer IP address to the DC01



- Access Denied



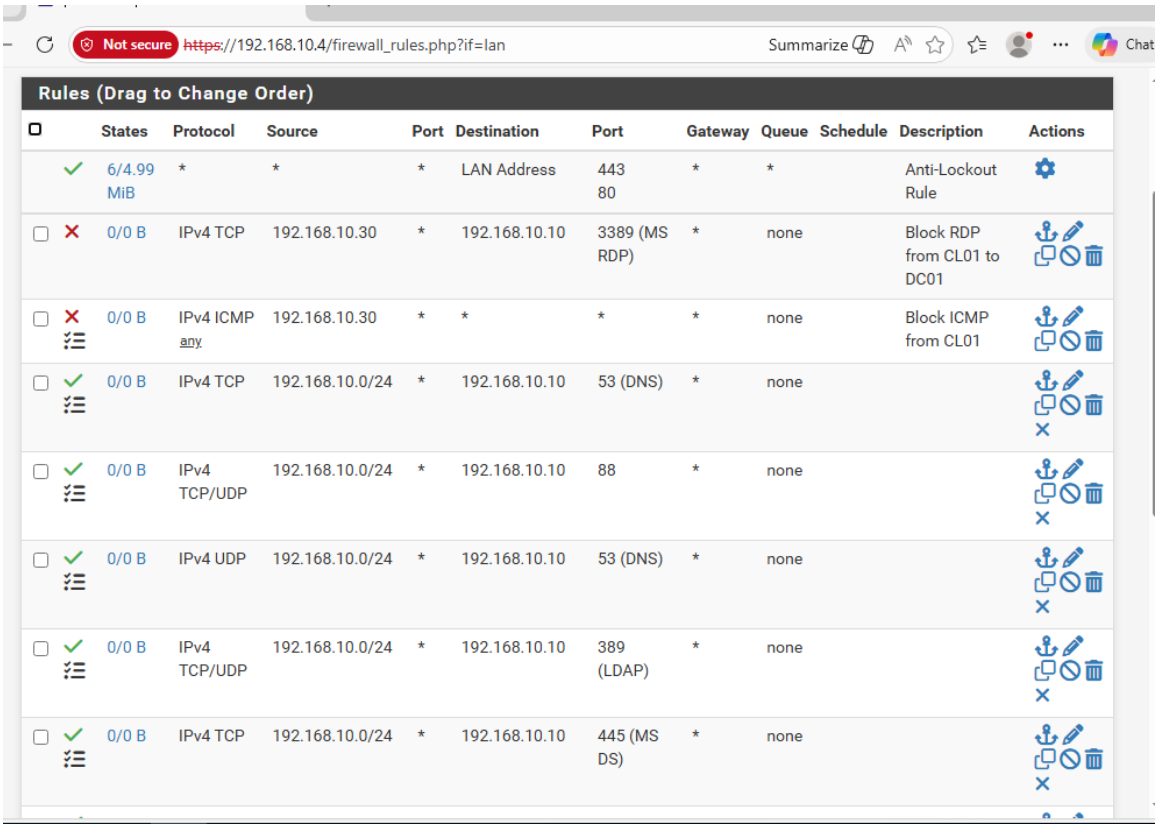
5.3 Rule 3 — Allow Active Directory Services to DC01

Pass rules added for all required Active Directory ports from internal network to DC01:

Port	Protocol	Service	Description
53	TCP	DNS	Allow-DNS-TCP-to-DC01
53	UDP	DNS	Allow-DNS-UDP-to-DC01
88	TCP/UDP	Kerberos	Allow-Kerberos-to-DC01
389	TCP/UDP	LDAP	Allow-LDAP-to-DC01
445	TCP	SMB	Allow-SMB-to-DC01
636	TCP	LDAPS	Allow-LDAPS-to-DC01
3268	TCP	Global Catalog	Allow-GlobalCatalog-to-DC01
3269	TCP	Global Catalog SSL	Allow-GlobalCatalogSSL-to-DC01

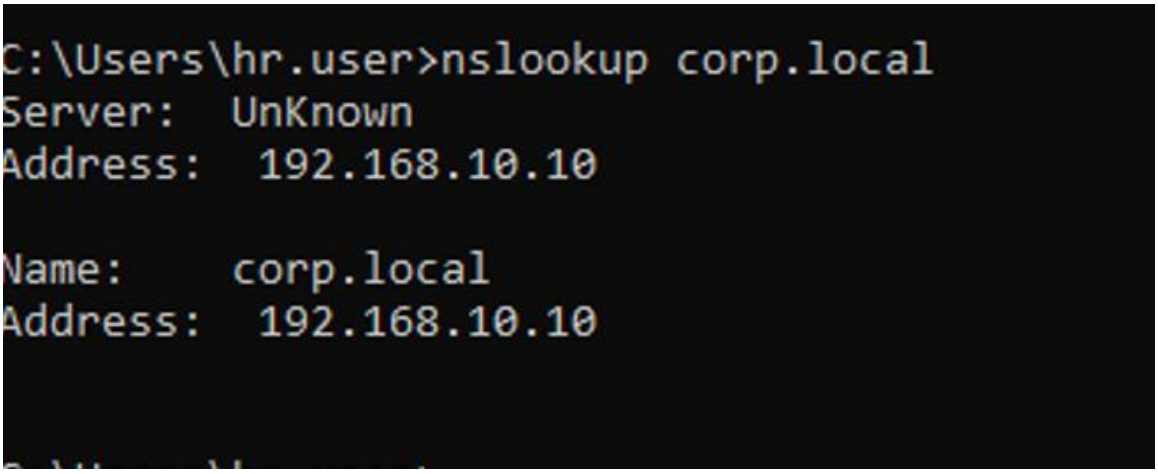
Test Result: nslookup corp.local on CL01 returned 192.168.10.10 — DNS (port 53) allowed through pfSense correctly.

- LAN rules for allow and block, block rules on the top so they will be checked first



States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
6/4.99 MiB	*	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
0/0 B	IPv4 TCP	192.168.10.30	*	192.168.10.10	3389 (MS RDP)	*	none		Block RDP from CL01 to DC01	
0/0 B	IPv4 ICMP any	192.168.10.30	*	*	*	*	none		Block ICMP from CL01	
0/0 B	IPv4 TCP	192.168.10.0/24	*	192.168.10.10	53 (DNS)	*	none			
0/0 B	IPv4 TCP/UDP	192.168.10.0/24	*	192.168.10.10	88	*	none			
0/0 B	IPv4 UDP	192.168.10.0/24	*	192.168.10.10	53 (DNS)	*	none			
0/0 B	IPv4 TCP/UDP	192.168.10.0/24	*	192.168.10.10	389 (LDAP)	*	none			
0/0 B	IPv4 TCP	192.168.10.0/24	*	192.168.10.10	445 (MS DS)	*	none			

- Enter the nslookup command to check for the DC01



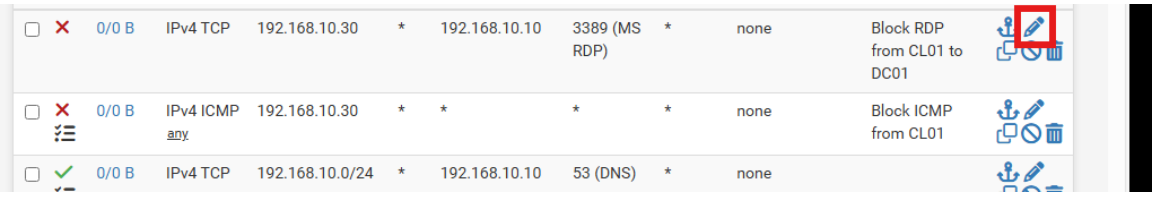
5.4 Rule 4 — Allow VPN Client Traffic

Setting	Value
Action	Pass
Protocol	Any

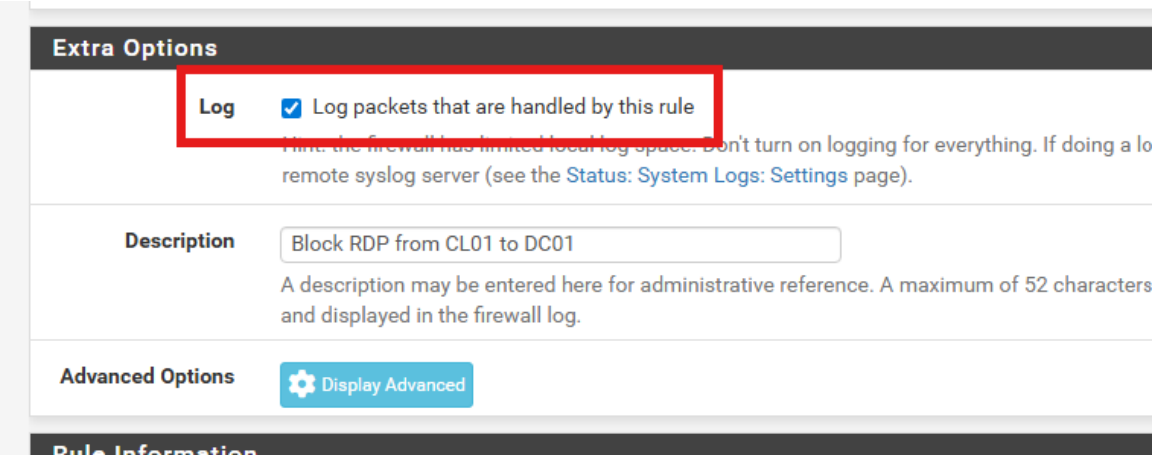
Setting	Value
Source	192.168.30.0/24 (VPN clients)
Destination	192.168.10.0/24 (LAN)
Description	Allow-VPN-Clients-to-LAN

5.5 Logging Enabled on All Rules

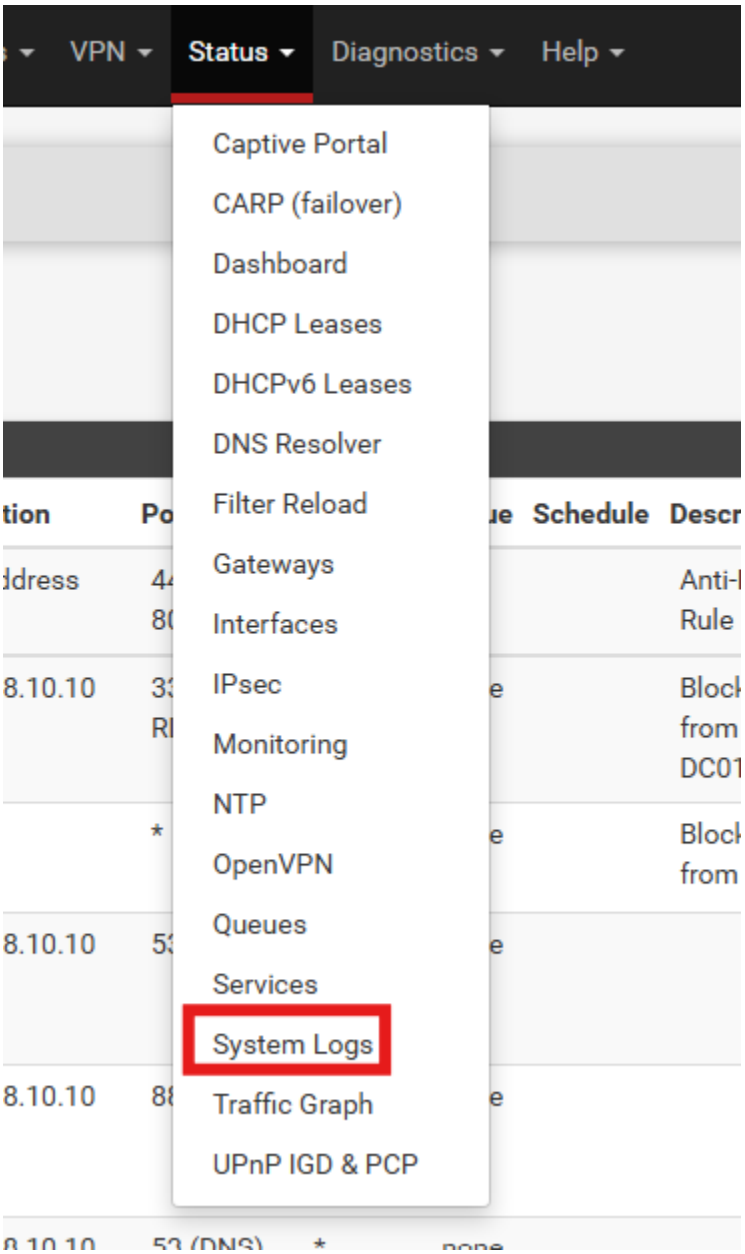
Logging was enabled on every firewall rule (edit rule > check Log packets that are handled by this rule > Save > Apply Changes). Firewall logs viewable at Status > System Logs > Firewall tab and via dynamic view at status_logs_filter_dynamic.php.



- Check the “Log Packet that are handled by this rule” option



- Go to Status → System Logs

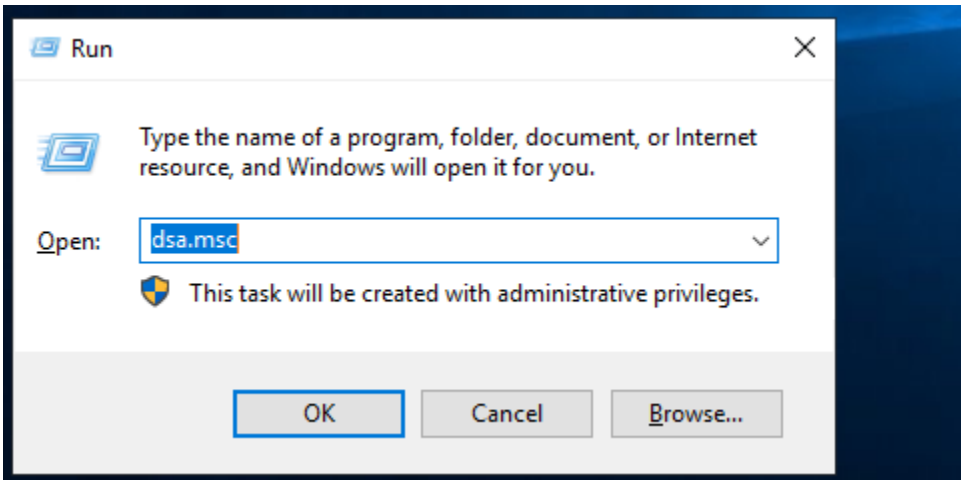


6. Active Directory Verification

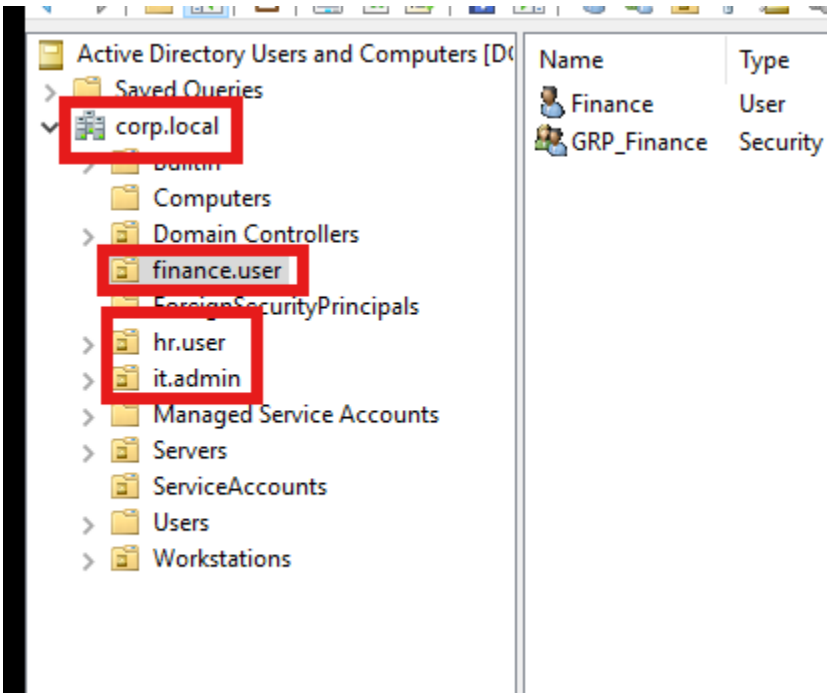
After the gateway change, Active Directory integrity was verified to confirm Modules 2 and 3 configurations remained intact.

6.1 OU and User Verification (DC01)

Run: Win+R > dsa.msc — confirmed the following OUs exist in corp.local:



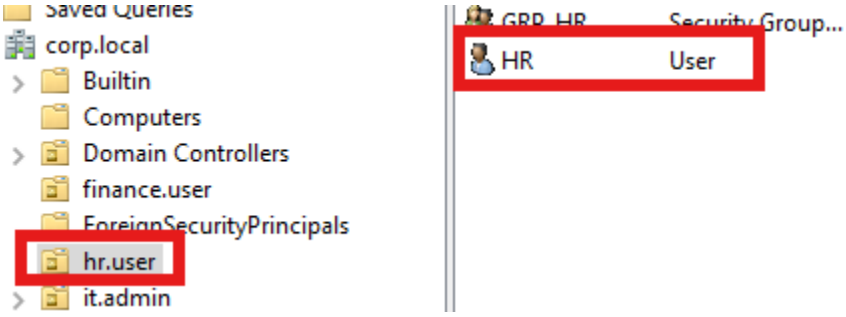
- Go to corp.local anc check if the finance, hr and admin groups exist



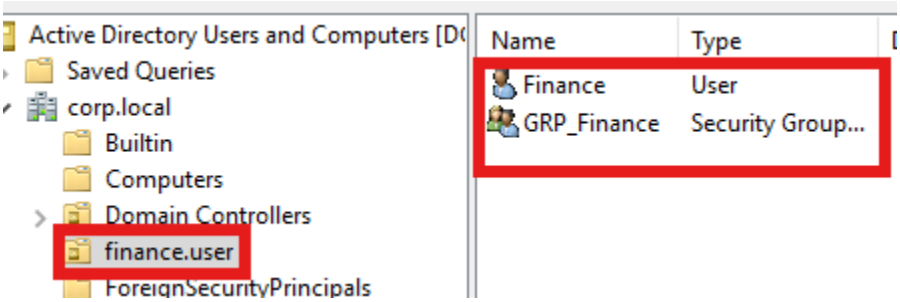
- IT, HR, Finance, Workstations, Servers, ServiceAccounts

6.2 Group Membership Checks

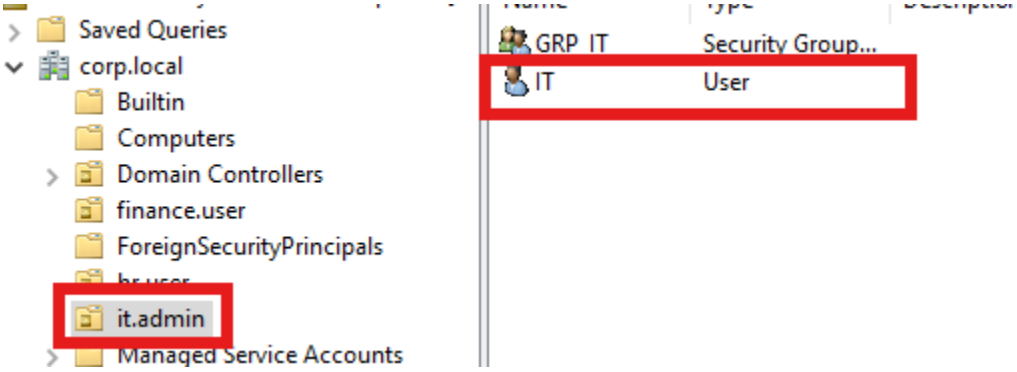
- GRP_HR: hr.user confirmed as member
- GRP_Finance: finance.user confirmed as member
- GRP_IT: it.admin confirmed as member



- Finance Ou users



- IT Admin OU



6.3 GPO Force Update

Ran on DC01, ADMIN01, and CL01 in PowerShell (Administrator):

```
gpupdate /force
```

```
PS C:\Users\Administrator> gpupdate /force
Updating policy...

Computer Policy update has completed successfully.
User Policy update has completed successfully.

PS C:\Users\Administrator>
```

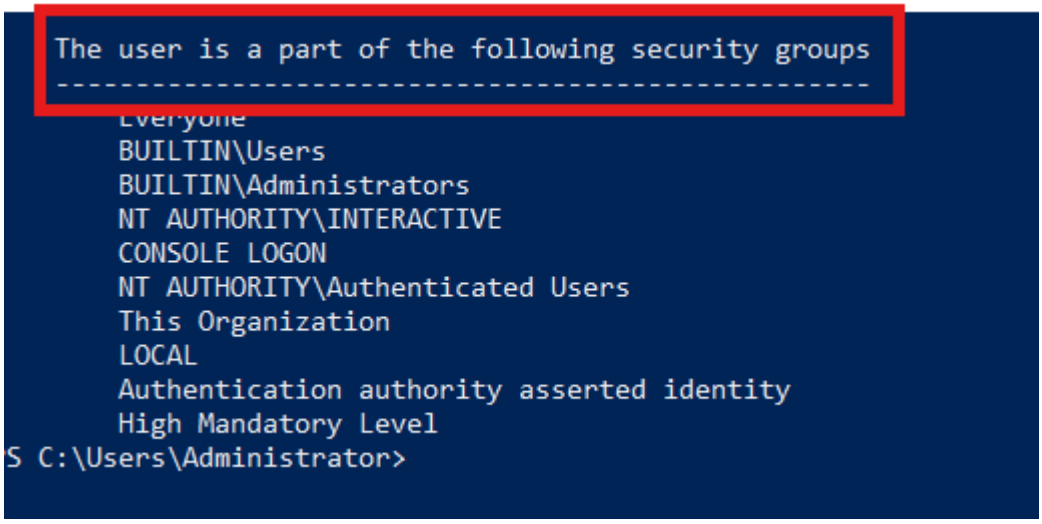
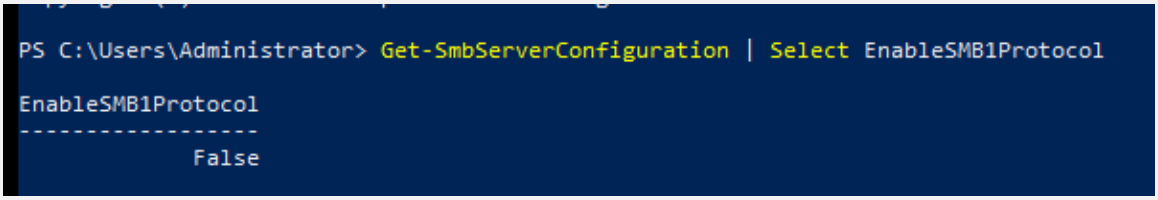
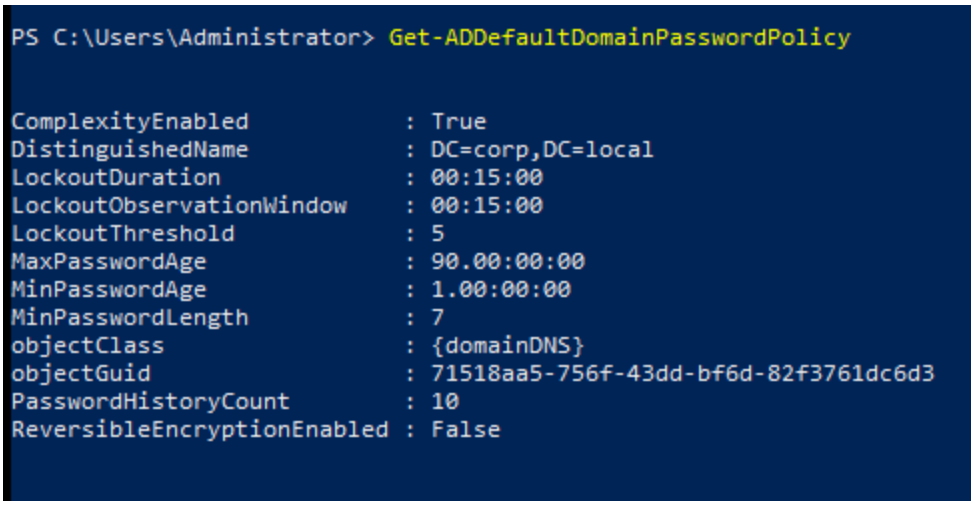
- GPU rules successfully updated

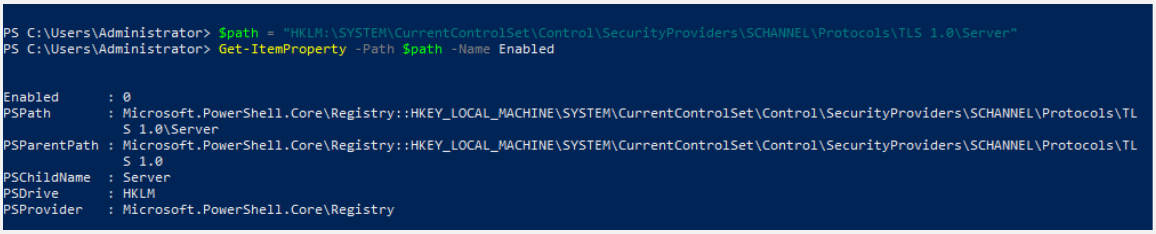
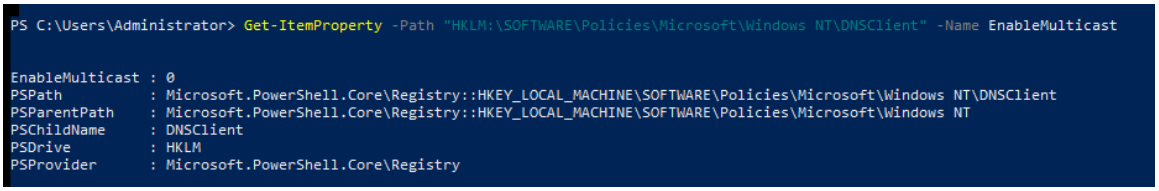
7. GPO Verification After Gateway Change

All Group Policy Objects from Module 3 were verified to still be applying correctly after the pfSense gateway change.

GPO Check	Command Used		Expecte
GPO applying on CL01	gpresult /r (as hr.user)	Applied Group Policy Objects listed	

```
PS C:\Users\Administrator> gpresult /r

Microsoft (R) Windows (R) Operating System Group Policy Result tool v2.0
© Microsoft Corporation. All rights reserved.

Created on ⌈5/⌈1/⌈2026 at 6:32:00 AM


RSOP data for CORP\Administrator on CL01 : Logging Mode
-------------------------------------------------------

OS Configuration:            Member Workstation
OS Version:                  10.0.19045
Site Name:                   Default-First-Site-Name
Roaming Profile:             N/A
Local Profile:               C:\Users\Administrator
Connected over a slow link?: No


COMPUTER SETTINGS
-----------------

    Last time Group Policy was applied: 5/1/2026 at 5:27:26 AM
    Group Policy was applied from:      DC01.corp.local
    Group Policy slow link threshold:   500 kbps
    Domain Name:                        CORP
    Domain Type:                        WindowsNT 4

    Applied Group Policy Objects
    ----------------------------
        Default Domain Policy
```

GPO Check	Command Used	Expecte
		
SMBv1 Disabled	Get-SmbServerConfiguration | Select EnableSMB1Protocol 	False
Account Lockout Active	Get-ADDefaultDomainPasswordPolicy 	Lockout t
TLS 1.0 Disabled	Get-ItemProperty (SCHANNEL TLS 1.0 path) -Name Enabled	Enabled:

GPO Check	Command Used	Expecte
	 PS C:\Users\Administrator> $path = "HKLM:\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.0\Server" PS C:\Users\Administrator> Get-ItemProperty -Path $path -Name Enabled Enabled : 0 PSPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.0\Server PSParentPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\SecurityProviders\SCHANNEL\Protocols\TLS 1.0 PSChildName : Server PSDrive : HKLM PSProvider : Microsoft.PowerShell.Core\Registry	
LLMNR Disabled	Get-ItemProperty (DNSClient path) -Name EnableMulticast  PS C:\Users\Administrator> Get-ItemProperty -Path "HKLM:\SOFTWARE\Policies\Microsoft\Windows NT\DNSClient" -Name EnableMulticast EnableMulticast : 0 PSPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows NT\DNSClient PSParentPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Policies\Microsoft\Windows NT PSChildName : DNSClient PSDrive : HKLM PSProvider : Microsoft.PowerShell.Core\Registry	0

8. Suricata IDS Rules Update

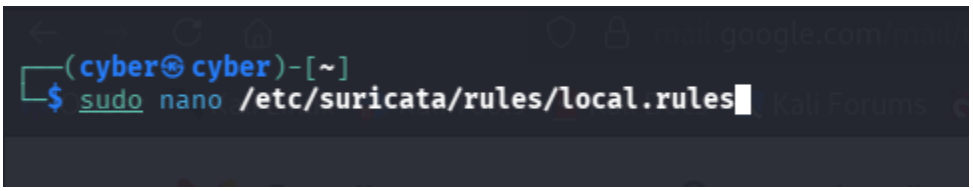
Suricata was confirmed running on IDS01 (Kali Linux, 192.168.10.60). Custom detection rules were added to /etc/suricata/rules/local.rules.

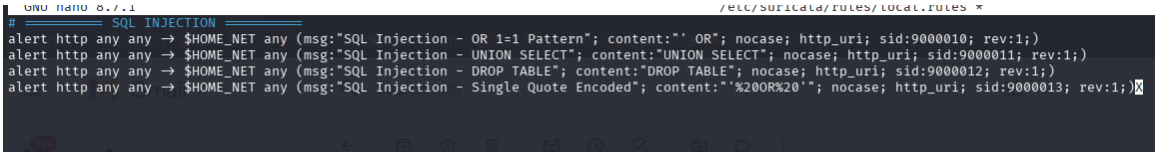
8.1 SQL Injection Detection Rules

- Suricata Custom SQL Injection Rules

```
alert http any any -> $HOME_NET any (msg:"SQL Injection - OR 1=1 Pattern"; content:"' OR"; nocase; http_uri; sid:9000010; rev:1;)
alert http any any -> $HOME_NET any (msg:"SQL Injection - UNION SELECT"; content:"UNION SELECT"; nocase; http_uri; sid:9000011; rev:1;)
alert http any any -> $HOME_NET any (msg:"SQL Injection - DROP TABLE"; content:"DROP TABLE"; nocase; http_uri; sid:9000012; rev:1;)
alert http any any -> $HOME_NET any (msg:"SQL Injection - Single Quote Encoded"; content:"'%20OR%20'"; nocase; http_uri; sid:9000013; rev:1;)
```

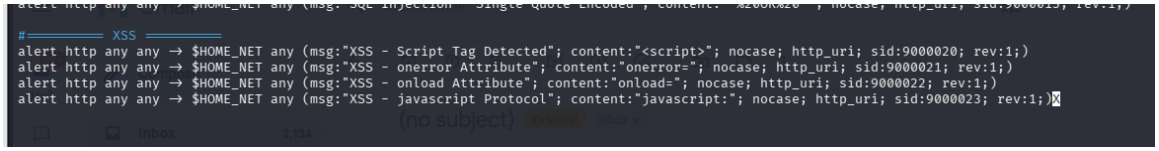
- Editing Suricata Local.rules file





8.2 XSS Detection Rules

```
alert http any any -> $HOME_NET any (msg:"XSS - Script Tag Detected"; content:"<script>"; nocase; http_uri; sid:9000020; rev:1;)
alert http any any -> $HOME_NET any (msg:"XSS - onerror Attribute"; content:"onerror="; nocase; http_uri; sid:9000021; rev:1;)
alert http any any -> $HOME_NET any (msg:"XSS - onload Attribute"; content:"onload="; nocase; http_uri; sid:9000022; rev:1;)
alert http any any -> $HOME_NET any (msg:"XSS - javascript Protocol"; content:"javascript:"; nocase; http_uri; sid:9000023; rev:1;)
```



8.3 Brute Force Detection Rules

- Adding Brute force rules

```
alert tcp any any -> $HOME_NET 3389 (msg:"Brute Force - RDP Login Attempts"; threshold:type threshold, track by_src, count 10, seconds 60; sid:9000030; rev:1;)
alert tcp any any -> $HOME_NET 22 (msg:"Brute Force - SSH Login Attempts"; threshold:type threshold, track by_src, count 10, seconds 60; sid:9000031; rev:1;)
alert tcp any any -> $HOME_NET 445 (msg:"Brute Force - SMB Authentication"; threshold:type threshold, track by_src, count 5, seconds 30; sid:9000032; rev:1;)
```

```
# ========== BRUTE FORCE ==========
alert tcp any any -> $HOME_NET 3389 (msg:"Brute Force - RDP Login Attempts"; threshold:type threshold, track by_src, count 10, seconds 60; sid:9000030; rev:1;)
alert tcp any any -> $HOME_NET 22 (msg:"Brute Force - SSH Login Attempts"; threshold:type threshold, track by_src, count 10, seconds 60; sid:9000031; rev:1;)
alert tcp any any -> $HOME_NET 445 (msg:"Brute Force - SMB Authentication"; threshold:type threshold, track by_src, count 5, seconds 30; sid:9000032; rev:1;)
```

8.4 LLMNR / NBT-NS Poisoning Detection Rules

- LLMNR Poisoning detection Rules

```
alert udp any any -> 224.0.0.252 5355 (msg:"LLMNR Multicast Query - Possible Poisoning"; sid:9000040; rev:1;)
alert udp any any -> $HOME_NET 5355 (msg:"LLMNR Unicast Query Detected"; sid:9000041; rev:1;)
alert udp any any -> $HOME_NET 137 (msg:"NBT-NS Query Detected - Possible Poisoning"; sid:9000042; rev:1;)
alert udp any any -> $HOME_NET 138 (msg:"NetBIOS Datagram Service Traffic Detected"; sid:9000043; rev:1;)
```

- Custom Rules for LLMNR attack detection

```
# ========== LLMNR / NBT-NS ==========
alert udp any any → 224.0.0.252 5355 (msg:"LLMNR Multicast Query - Possible Poisoning"; sid:9000040; rev:1;)
alert udp any any → $HOME_NET 5355 (msg:"LLMNR Unicast Query Detected"; sid:9000041; rev:1;)
alert udp any any → $HOME_NET 137 (msg:"NBT-NS Query Detected - Possible Poisoning"; sid:9000042; rev:1;)
alert udp any any → $HOME_NET 138 (msg:"NetBIOS Datagram Service Traffic Detected"; sid:9000043; rev:1;)
```

8.5 Suricata Restart and Verification

```
sudo suricata -T -c /etc/suricata/suricata.yaml -v   # Syntax test
sudo systemctl restart suricata
sudo systemctl status suricata                        # Confirmed:
active (running)
sudo tail -f /var/log/suricata/fast.log               # Live alert
monitor
```

- Saving and testing the configuration to check for any errors in the local.rules

```
┌──(cyber㉿cyber)-[~]
└─$ sudo suricata -T -c /etc/suricata/suricata.yaml -v
Notice: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Running suricata under test mode
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log

Info: detect: 2 rule files processed. 49108 rules successfully loaded, 0 rules failed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 49113 signatures processed. 1270 are IP-only rules, 4477 are inspecting packet payload, 43130 inspect application layer, 1
Notice: mpm-hs: Rule group caching - loaded: 113 newly cached: 0 total cacheable: 113
Notice: suricata: Configuration provided was successfully loaded. Exiting.
```

- Checking for Suricata status

```
┌──(cyber㉿cyber)-[~]
└─$ sudo systemctl restart suricata

┌──(cyber㉿cyber)-[~]
└─$ sudo systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
     Loaded: loaded (/usr/lib/systemd/system/suricata.service; enabled; preset: disabled)
     Active: active (running) since Sat 2026-05-02 08:46:29 EDT; 37s ago
 Invocation: 4908361b5c4547c3b957c7eb731b9d01
       Docs: man:suricata(8)
             man:suricatasc(8)
             https://suricata.io/documentation/
   Main PID: 72861 (Suricata-Main)
      Tasks: 8 (limit: 4414)
     Memory: 452.9M (peak: 452.9M)
        CPU: 22.886s
     CGroup: /system.slice/suricata.service
             └─72861 /usr/bin/suricata -D --af-packet -c /etc/suricata/suricata.yaml --pidfile /var/run/suricata/surica

May 02 08:46:06 cyber systemd[1]: Starting suricata.service - Suricata IDS/IDP daemon...
May 02 08:46:06 cyber suricata[72860]: i: suricata: This is Suricata version 8.0.3 RELEASE running in SYSTEM mode
May 02 08:46:29 cyber systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
```

- Checking for the fast.logs file for detection logs

```
┌──(cyber㉿cyber)-[~]
└─$ sudo tail -f /var/log/suricata/fast.log
05/02/2026-08:27:57.408192  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound like
.60:38776 -> 104.17.254.239:80
05/02/2026-08:27:57.423624  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound like
.60:38776 -> 104.17.254.239:80
05/02/2026-08:27:58.781447  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound like
.60:38776 -> 104.17.254.239:80
05/02/2026-08:27:58.994063  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound like
.60:38776 -> 104.17.254.239:80
05/02/2026-08:27:59.095129  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound like
.60:38776 -> 104.17.254.239:80
05/02/2026-08:28:01.179855  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outbound like
.60:58360 -> 202.28.95.174:80
05/02/2026-08:28:05.188048  [**] [1:9000043:1] NetBIOS Datagram Service Traffic Detected [**]
05/02/2026-08:30:53.064824  [**] [1:9000040:1] LLMNR Multicast Query - Possible Poisoning [**
05/02/2026-08:30:53.912218  [**] [1:9000040:1] LLMNR Multicast Query - Possible Poisoning [**
```

Note: During setup, DNS resolution on IDS01 failed due to /etc/resolv.conf not being set. Fixed by running:

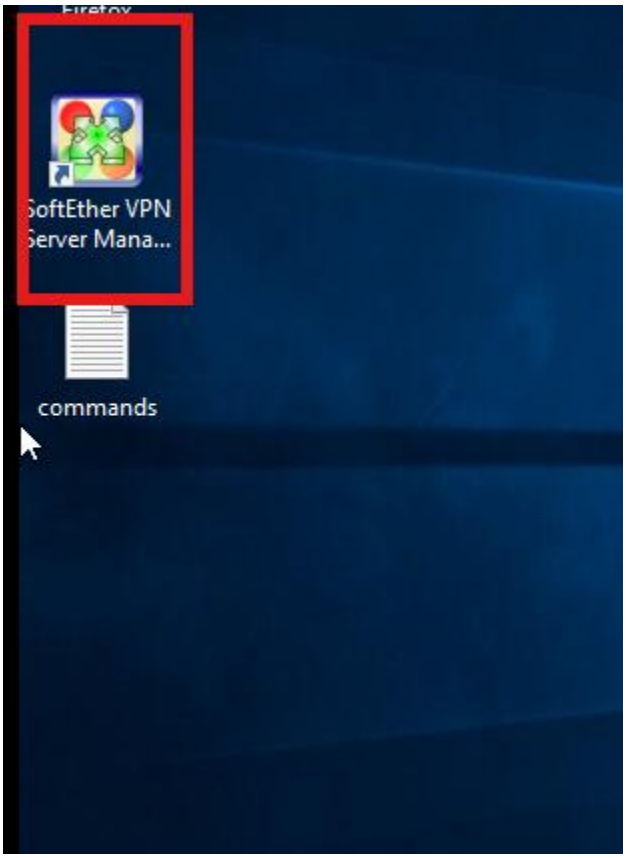
```
echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
```

9. VPN Verification

9.1 VPN01 SoftEther Status

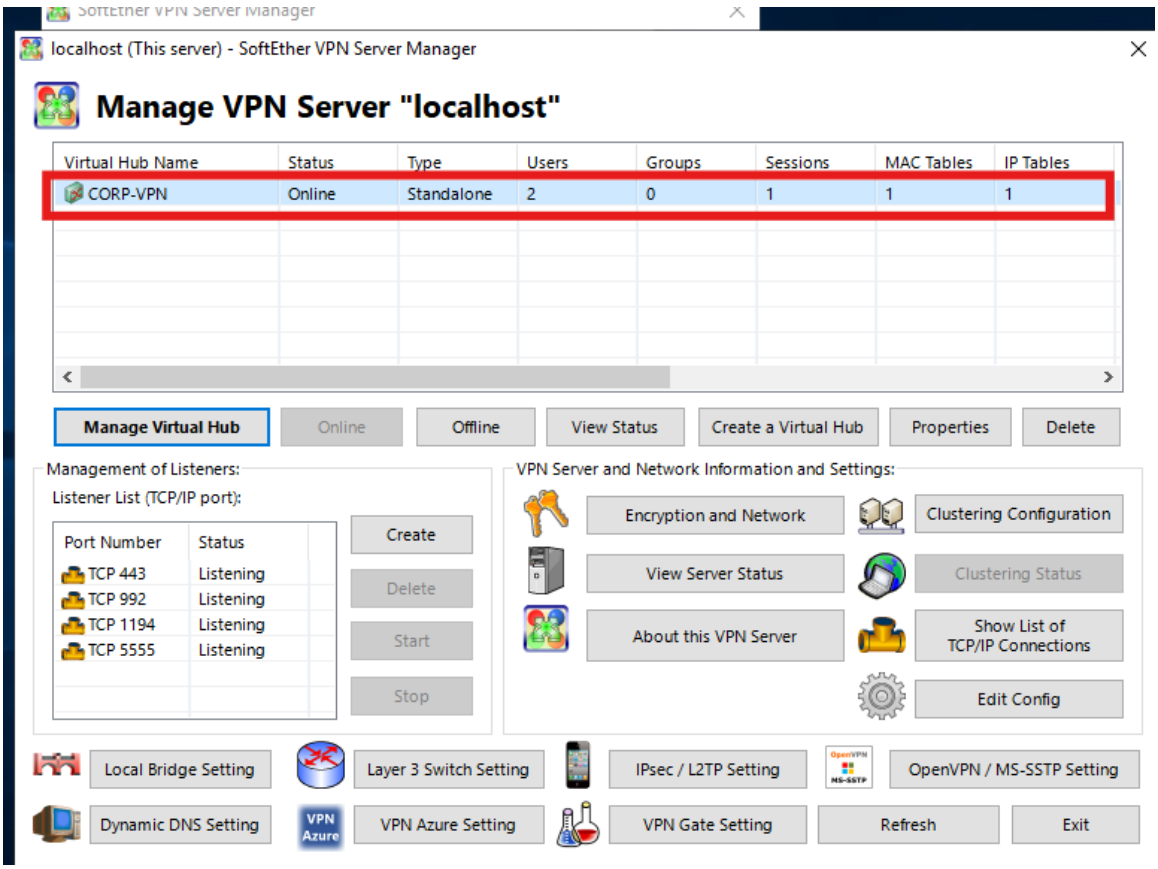
SoftEther VPN Server Manager on VPN01 confirmed:

- CORP-VPN hub: Online
- Hub Type: Standalone
- Sessions: 1 active
- Listeners active: TCP 443, 992, 1194, 5555



- Starting and connecting the SoftEther VPN

- Checking for the status of the VPN, On or Off



9.2 L2TP/IPSec Client Setup on ATTACK01

Required packages were installed on ATTACK01 (Kali Linux) to support L2TP/IPSec VPN:

```
sudo apt install strongswan xl2tpd network-manager-l2tp network-manager-l2tp-gnome -y
sudo systemctl start xl2tpd
sudo systemctl restart NetworkManager
```

VPN connection profile created with these settings:

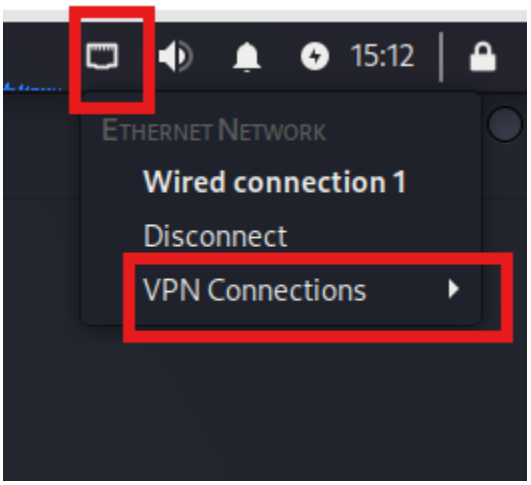
- Connection Name: CORP-VPN-Test
- Gateway: 192.168.10.50 (VPN01)
- Username: vpnuser | Password: Admin123
- IPSec Pre-shared key: Corp@IPSec2024

```
┌──(cyber㉿cyber)-[~]
└─$ which l2tpd && echo installed || echo missing
l2tpd not found
missing
```

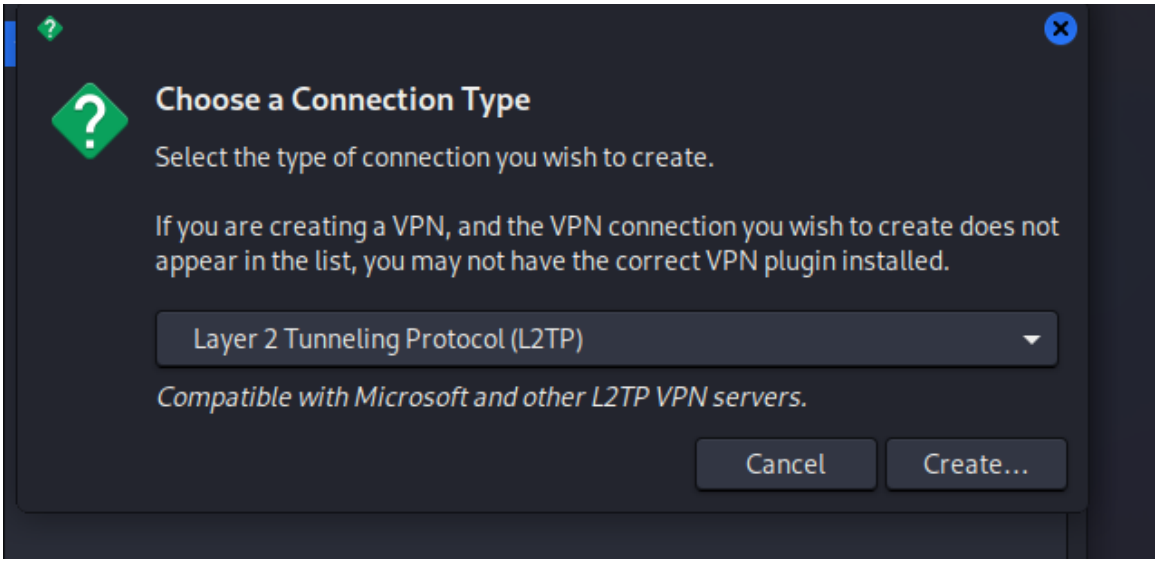
- VPN connection file creation

```
┌──(cyber㉿cyber)-[~]
└─$ sudo apt install network-manager-l2tp network-manager-l2tp-gnome -y
network-manager-l2tp is already the newest version (1.52.0-2).
network-manager-l2tp set to manually installed.
network-manager-l2tp-gnome is already the newest version (1.52.0-2).
network-manager-l2tp-gnome set to manually installed.
The following packages were automatically installed and are no longer required:
  libcrypt-dev  libgdk-pixbuf2.0-bin  libplacebo351  librubberband2  libsframe2  libsnmp40t64  linux-image-6.17.10+kali-amd64
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 275
```

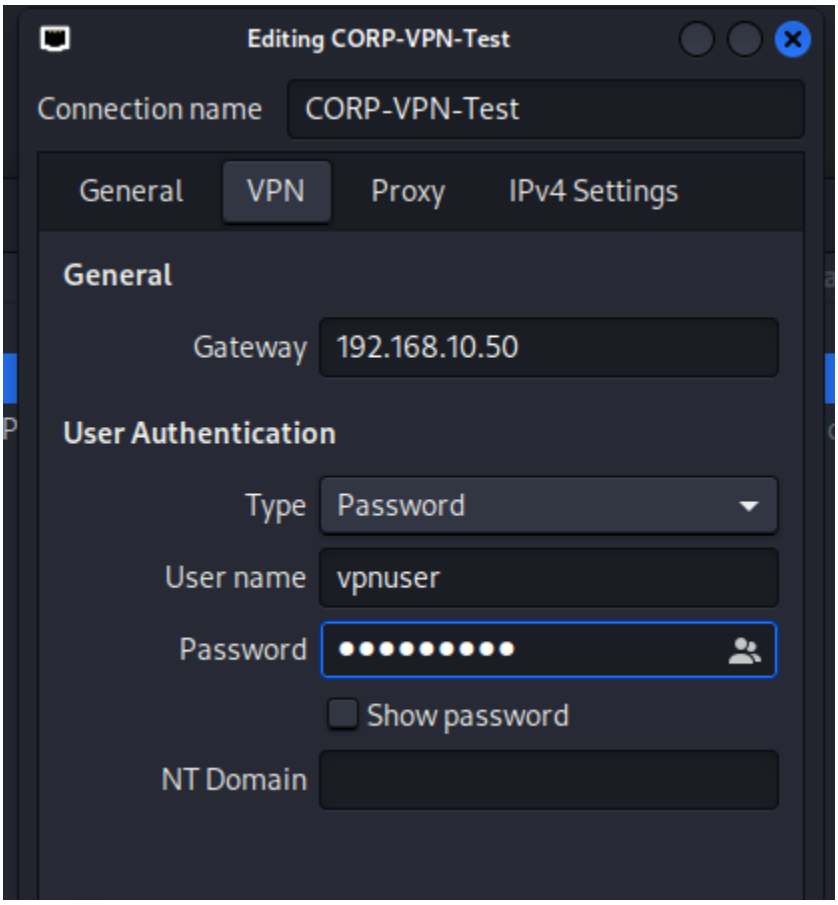
- Connecting with the VPN profile



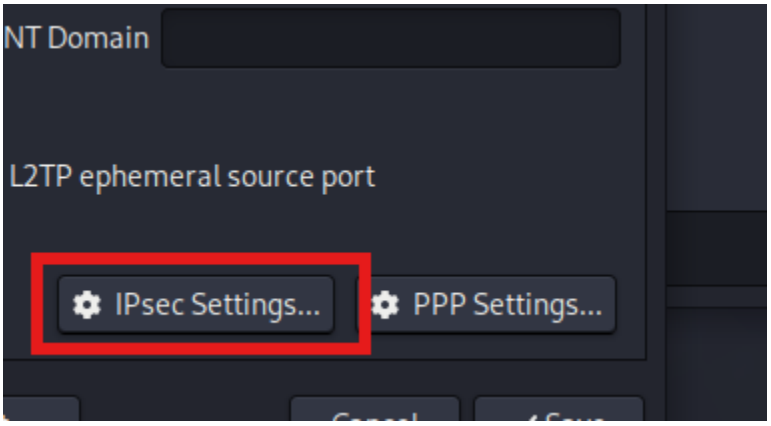
- Select L2TP



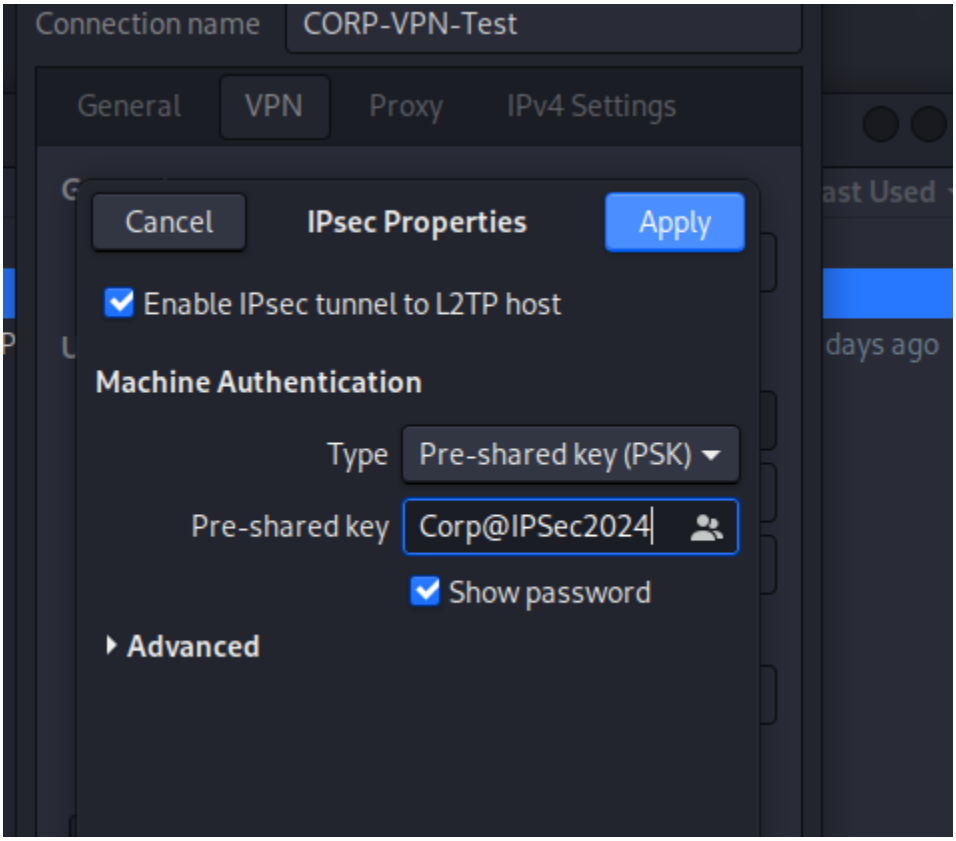
- Enter the Gateway as the Windows Server VPN



- Click on IPsec Settings

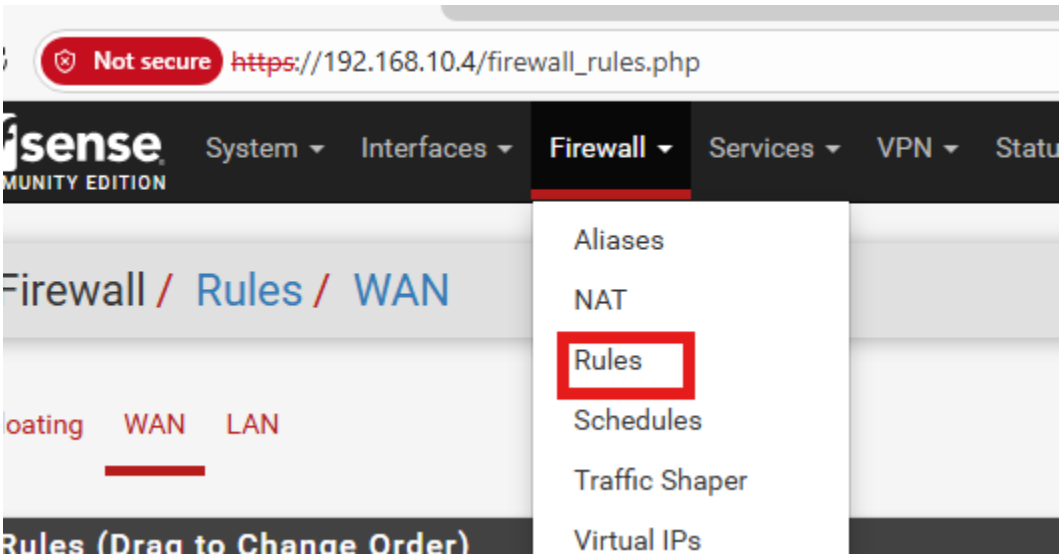


- Enter the PreShared key

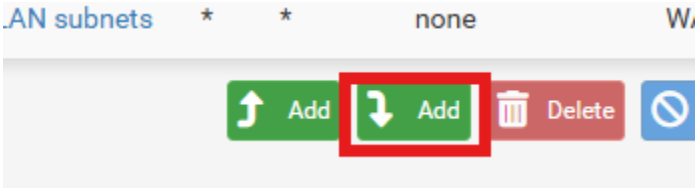


9.3 pfSense VPN Pass Rule

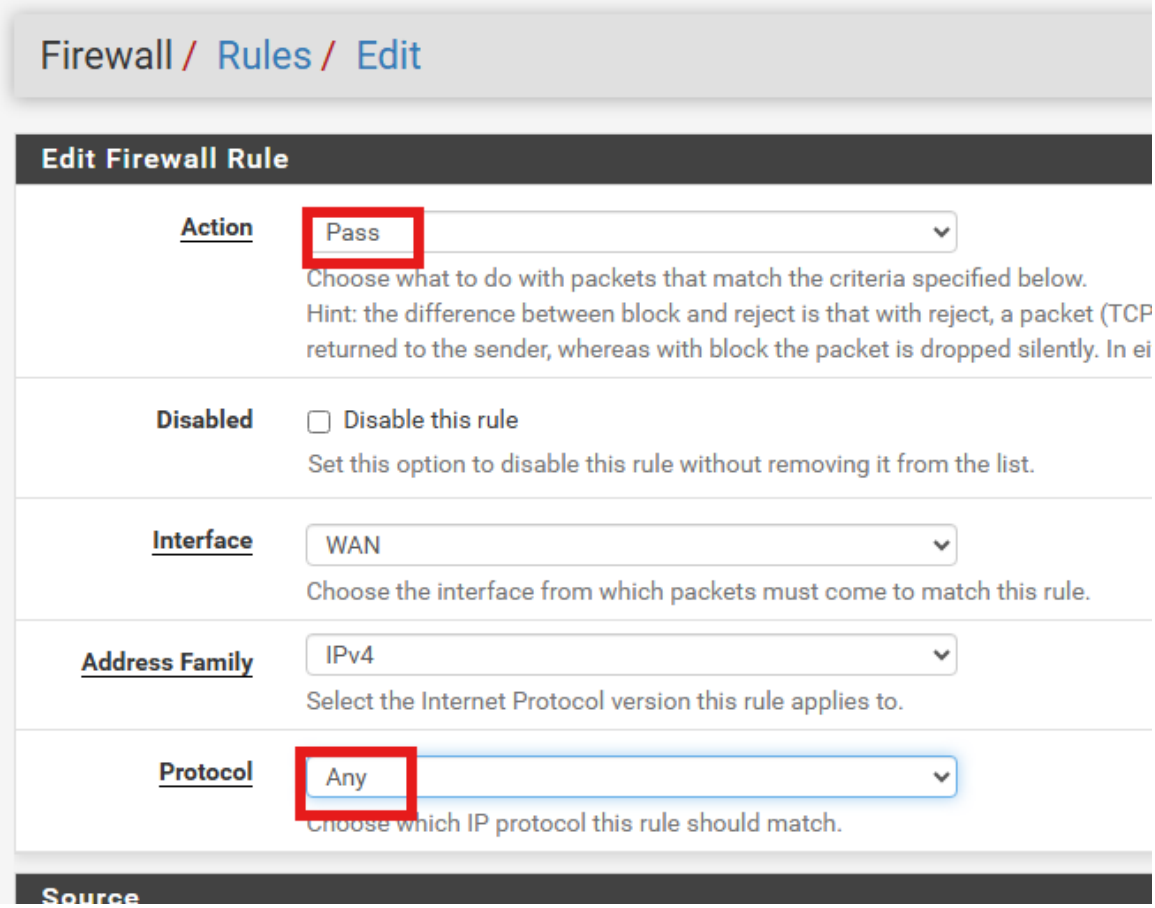
A pass rule was added in pfSense to allow VPN client traffic (192.168.30.0/24) to reach the internal LAN (192.168.10.0/24). Description: Allow-VPN-Clients-to-LAN.



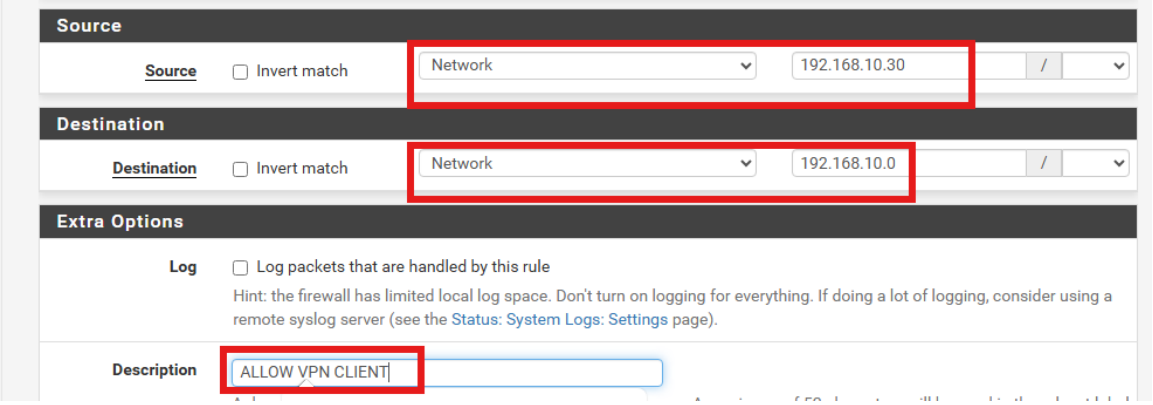
Add the Pass rule for the Firewall, below the blocking rules



- Set the Action to Allow and select the Protocol to “Any”



- Set the Source to a Network and Destination to the Network as well



10. Attack Simulation

> **All attacks were executed from ATTACK01 (192.168.10.40). IDS01 live monitor (tail -f fast.log) and pfSense Firewall logs were monitored simultaneously during all attacks.**

10.1 Attack 1 — Network Reconnaissance

Goal: Discover live hosts, open ports, OS, and service versions.

```
nmap -A -T5 192.168.10.10           # Aggressive scan DC01
nmap -sS -p 1-1024 192.168.10.10    # Port scan DC01
nmap -A -T4 192.168.10.10            # OS and version detection
nmap -sU -p 500,1701,4500 192.168.10.50  # VPN server UDP ports
```

Expected: Open ports discovered on DC01. Suricata and pfSense log the scan activity.

```
┌──(raphael㉿raphael)-[~]
└─$ nmap -A -T5 192.168.10.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-01 16:01 -0400
Stats: 0:00:09 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 14.29% done; ETC: 16:02 (0:00:36 remaining)
Stats: 0:01:11 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.95% done; ETC: 16:02 (0:00:00 remaining)
Nmap scan report for 192.168.10.10
Host is up (0.00087s latency).
Not shown: 986 filtered tcp ports (no-response)
PORT     STATE SERVICE       VERSION
22/tcp   open  ssh           OpenSSH for_Windows_7.7 (protocol 2.0)
| ssh-hostkey:
|   2048 18:d4:ee:68:0b:13:75:ae:72:7b:02:32:11:f6:7b:35 (RSA)
|   256 62:09:22:06:f9:e1:84:a8:5e:1d:37:96:ee:a9:4f:78 (ECDSA)
|_  256 8a:6f:6f:be:b6:70:e8:29:b8:6c:0f:d4:be:a4:dd:61 (ED25519)
53/tcp   open  domain        Simple DNS Plus
80/tcp   open  http          Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
|_http-title: IIS Windows Server
| http-methods:
|_  Potentially risky methods: TRACE
88/tcp   open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-05-01 20:01:25Z)
135/tcp  open  msrpc         Microsoft Windows RPC
389/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: corp.local, Site: Defaul
| ssl-cert: Subject:
| Subject Alternative Name: DNS:DC01.corp.local
```

- Nmap port and services Scan

```
┌──(raphael㉿raphael)-[~]
└─$ nmap -sS -p 1-1024 192.168.10.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-01 16:03 -0400
Nmap scan report for 192.168.10.10
Host is up (0.00083s latency).
Not shown: 1014 filtered tcp ports (no-response)
PORT    STATE SERVICE
22/tcp  open  ssh
53/tcp  open  domain
80/tcp  open  http
88/tcp  open  kerberos-sec
135/tcp open  msrpc
389/tcp open  ldap
445/tcp open  microsoft-ds
464/tcp open  kpasswd5
593/tcp open  http-rpc-epmap
636/tcp open  ldapssl
MAC Address: 00:0C:29:D0:E8:38 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 5.40 seconds

┌──(raphael㉿raphael)-[~]
```

- Nmap scan that creates a lot of noise in the network

```
┌──(raphael㉿raphael)-[~]
└─$ nmap -A -T4 192.168.10.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-01 16:03 -0400
Stats: 0:00:24 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 85.71% done; ETC: 16:04 (0:00:03 remaining)
Stats: 0:00:53 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.95% done; ETC: 16:04 (0:00:00 remaining)
Stats: 0:01:10 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 99.95% done; ETC: 16:04 (0:00:00 remaining)
Nmap scan report for 192.168.10.10
Host is up (0.00089s latency).
Not shown: 986 filtered tcp ports (no-response)
PORT     STATE SERVICE       VERSION
22/tcp   open  ssh           OpenSSH for_Windows_7.7 (protocol 2.0)
| ssh-hostkey:
|   2048 18:d4:ee:68:0b:13:75:ae:72:7b:02:32:11:f6:7b:35 (RSA)
|   256 62:09:22:06:f9:e1:84:a8:5e:1d:37:96:ee:a9:4f:78 (ECDSA)
|_  256 8a:6f:6f:be:b6:70:e8:29:b8:6c:0f:d4:be:a4:dd:61 (ED25519)
53/tcp   open  domain        Simple DNS Plus
80/tcp   open  http          Microsoft IIS httpd 10.0
|_http-server-header: Microsoft-IIS/10.0
|_http-title: IIS Windows Server
| http-methods:
|_  Potentially risky methods: TRACE
88/tcp   open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-05-01 2
135/tcp  open  msrpc         Microsoft Windows RPC
389/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: corp
|_ssl-date: 2026-05-01T20:05:07+00:00; 0s from scanner time.
| ssl-cert: Subject:
```

10.2 Attack 2 — SMBv1 / EternalBlue Check

Goal: Confirm SMBv1 is disabled — this attack should FAIL, proving Module 3 hardening worked.

```
nmap --script smb-protocols 192.168.10.10
nmap --script smb-vuln-ms17-010 192.168.10.10
```

Expected Result: SMBv1 does NOT appear. No VULNERABLE section — Module 3 hardening confirmed.

```
┌──(raphael㉿raphael)-[~]
└─$ nmap -script smb-protocols 192.168.10.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-01 16:07 -0400
Nmap scan report for 192.168.10.10
Host is up (0.00076s latency).
Not shown: 986 filtered tcp ports (no-response)
PORT     STATE SERVICE
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
88/tcp   open  kerberos-sec
135/tcp  open  msrpc
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
3268/tcp open  globalcatLDAP
3269/tcp open  globalcatLDAPssl
5357/tcp open  wsdapi
5986/tcp open  wsmans
MAC Address: 00:0C:29:D0:E8:38 (VMware)

Host script results:
| smb-protocols:
|   dialects:
|     2.0.2
|     2.1
|     3.0
|     3.0.2
|_    3.1.1
```

- Executing a custom script through the nmap scan

```
┌──(raphael㉿raphael)-[~]
└─$ nmap -script  smb-vuln-ms17-010 192.168.10.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-01 16:10 -0400
Nmap scan report for 192.168.10.10
Host is up (0.0010s latency).
Not shown: 986 filtered tcp ports (no-response)
PORT     STATE SERVICE
22/tcp   open  ssh
53/tcp   open  domain
80/tcp   open  http
88/tcp   open  kerberos-sec
135/tcp  open  msrpc
389/tcp  open  ldap
445/tcp  open  microsoft-ds
464/tcp  open  kpasswd5
593/tcp  open  http-rpc-epmap
636/tcp  open  ldapssl
3268/tcp open  globalcatLDAP
3269/tcp open  globalcatLDAPssl
5357/tcp open  wsdapi
5986/tcp open  wsmans
MAC Address: 00:0C:29:D0:E8:38 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 6.44 seconds

┌──(raphael㉿raphael)-[~]
└─$
```

10.3 Attack 3 — Brute Force Attack

Goal: Trigger account lockout policy and IDS brute force alert.

```
nano mywordlist.txt   # Created password wordlist with 14 entries
hydra -l hr.user -P mywordlist.txt -t 1 smb2://192.168.10.10
```

Expected Result: Account locked after 5 failed attempts. Hydra cannot find credentials. IDS fires brute force alert on SMB port 445.

```
  GNU nano 8.7
12345
admin
abc
welcome
qwerty
dragon
```

- Creating a word list through nano command

```
┌──(raphael㉿raphael)-[~]
└─$ sudo nano word.txt

┌──(raphael㉿raphael)-[~]
└─$ hydra -l hr.user -P mywordlist.txt -t 1 smb2://192.168.10.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organiza
way).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-01 16:21:07
[WARNING] Workgroup was not specified, using "WORKGROUP"
[DATA] max 1 task per 1 server, overall 1 task, 14 login tries (l:1/p:14), ~14 tries per task
[DATA] attacking smb2://192.168.10.10:445/
1 of 1 target completed, 0 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-01 16:21:08

┌──(raphael㉿raphael)-[~]
```

10.4 Attack 4 — LLMNR Poisoning Test

Goal: Confirm LLMNR is disabled — Responder should capture nothing.

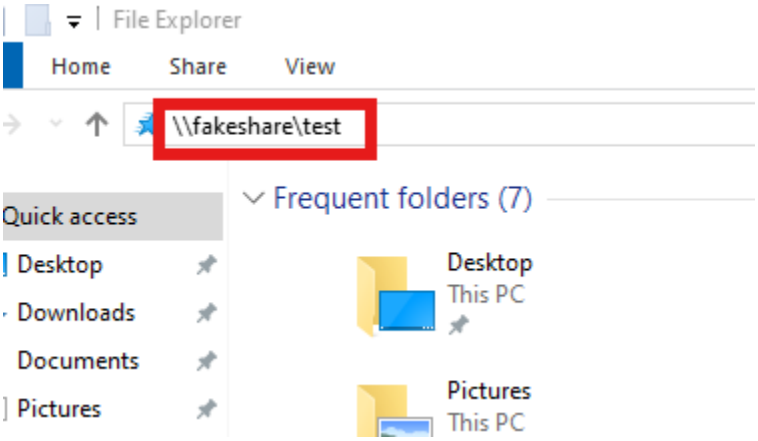
```
sudo responder -I eth0 -wv    # Started on ATTACK01
```

On CL01: File Explorer > address bar > \\fakeshare\test > Enter

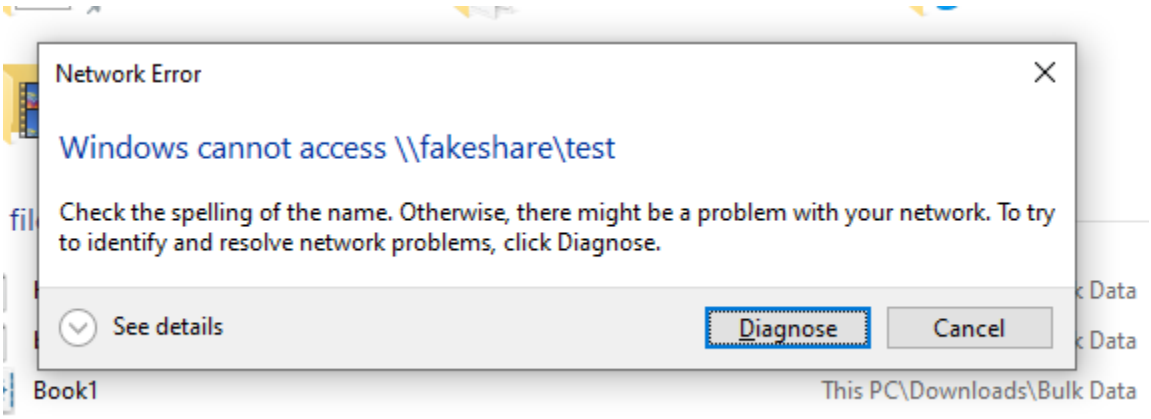
Expected Result: Responder captures no hashes because LLMNR GPO from Module 3 is still active.

```
┌──(raphael㉿raphael)-[~]
└─$ sudo responder -I eth0 -wv

[*] Tips jar:
    USDT → 0×Cc98c1D3b8cd9b717b5257827102940e4E17A19A
    BTC  → bc1q9360jedhhmps5vpl3u05vyg4jryrl52dmazz49

[+] Poisoners:
    LLMNR                      [ON]
    NBT-NS                     [ON]
    MDNS                       [ON]
    DNS                        [ON]
    DHCP                       [OFF]
    DHCPv6                     [OFF]

[+] Servers:
    HTTP server                [ON]
    HTTPS server               [ON]
    WPAD proxy                 [ON]
    Auth proxy                 [OFF]
    SMB server                 [ON]
    Kerberos server            [ON]
    SQL server                 [ON]
    FTP server                 [ON]
    IMAP server                [ON]
    POP3 server                [ON]
    SMTP server                [ON]
    DNS server                 [ON]
    LDAP server                [ON]
    MQTT server                [ON]
    RDP server                 [ON]
    DCE-RPC server             [ON]
    WinRM server               [ON]
    SNMP server                [ON]

[+] HTTP Options:
    Always serving EXE         [OFF]
```

- Checking if the LLMNR attack works



- Access denied LLMNR failed



10.5 Attack 5 — SQL Injection Test

Goal: Trigger Suricata SQL injection detection rules.

```
curl "http://192.168.10.10/Login.aspx" --data "username=' OR '1'='1&password=test"
curl "http://192.168.10.10/Login.aspx" --data "username=' UNION SELECT 1,2--&password=test"
```

Expected Result: Suricata fast.log shows SQL injection alerts (SID 9000010, 9000011).

```
┌──(raphael㉿raphael)-[~]
└─$ curl "http://192.168.10.10/Login.aspx" --data "username='OR'1'='1&password=test"


<html>
<head><title>Company Login</title></head>
<body>
    <h2>Login</h2>
    <form method="post">
        Username: <input type="text" name="username" /><br />
        Password: <input type="password" name="password" /><br />
        <input type="submit" value="Login" />
    </form>

<h3>Invalid password</h3>

</body>
</html>
```

- Testing SQL Injection attack

```
┌──(raphael㉿raphael)-[~]
└─$ curl "http://192.168.10.10/Login.aspx" --data "username='UNION SELECT 1,2--bpassword=test"
<!DOCTYPE html>
<html>
    <head>
        <title>All queries combined using a UNION, INTERSECT or EXCEPT operator must have an equal number of expressions in their target lists.</title>
        <meta name="viewport" content="width=device-width" />
        <style>
         body {font-family:"Verdana";font-weight:normal;font-size: .7em;color:black;}
         p {font-family:"Verdana";font-weight:normal;color:black;margin-top: -5px}
         b {font-family:"Verdana";font-weight:bold;color:black;margin-top: -5px}
         H1 { font-family:"Verdana";font-weight:normal;font-size:18pt;color:red }
         H2 { font-family:"Verdana";font-weight:normal;font-size:14pt;color:maroon }
         pre {font-family:"Consolas","Lucida Console",Monospace;font-size:11pt;margin:0;padding:0.5em;line-height:14pt}
         .marker {font-weight: bold; color: black;text-decoration: none;}
         .version {color: gray;}
         .error {margin-bottom: 10px;}
         .expandable { text-decoration:underline; font-weight:bold; color:navy; cursor:pointer; }
         @media screen and (max-width: 639px) {
          pre { width: 440px; overflow: auto; white-space: pre-wrap; word-wrap: break-word; }
         }
         @media screen and (max-width: 479px) {
          pre { width: 280px; }
         }
        </style>
    </head>

    <body bgcolor="white">

            <span><H1>Server Error in '/' Application.<hr width=100% size=1 color=silver></H1>

            <h2> <i>All queries combined using a UNION, INTERSECT or EXCEPT operator must have an equal number of expressions in their target lists.</i>

            <font face="Arial, Helvetica, Geneva, SunSans-Regular, sans-serif ">

            <b> Description: </b>An unhandled exception occurred during the execution of the current web request. Please review the stack trace for more
ed in the code.
```

10.6 Attack 6 — XSS Test

Goal: Trigger Suricata XSS detection rules.

```
curl "http://192.168.10.10/Comments.aspx" --data "comment=<script>alert(1)</script>"
curl "http://192.168.10.10/Comments.aspx" --data "comment=<img src=x onerror=alert(1)>"
```

Expected Result: Suricata fast.log shows XSS alerts (SID 9000020, 9000021).

```
┌──(raphael㉿raphael)-[~]
└─$ curl "http://192.168.10.10/Comments.aspx" --data "comment=<script>alert(1)</script>"


<!DOCTYPE html>
<html>
<head>
    <title>Comments Page</title>
</head>
<body>

<h2>Leave a Comment</h2>

<form method="post">
    Username:<br />
    <input type="text" name="username" /><br /><br />

    Comment:<br />
    <textarea name="comment" rows="4" cols="50"></textarea><br /><br />

    <input type="submit" value="Submit Comment" />
</form>

<hr />
<h3>All Comments:</h3>

<b>attacker</b>: <script>alert(1)</script><br /><br /><b>attacker</b>: <script>alert(1)</script><br /><b
)</script><br /><br /><b>attacker</b>: <script>alert('Network Security Module')</script><br /><br /><b>a
><br /><b></b>: <img src=x onerror=alert(1)><br /><br /><b></b>: <script>alert(1)</script><br /><br />

</body>
</html>
```

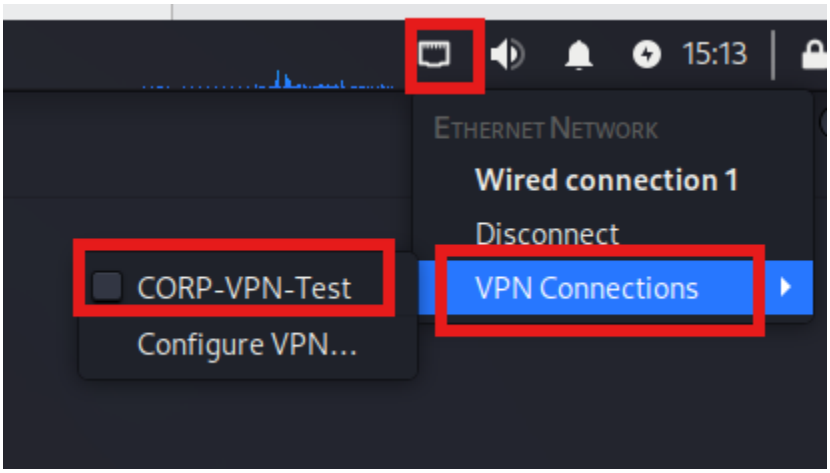
```
┌──(raphael㉿raphael)-[~]
└─$ curl "http://192.168.10.10/Comments.aspx" --data "comment=<img src=x onerroe=alert(1)"


<!DOCTYPE html>
<html>
<head>
    <title>Comments Page</title>
</head>
<body>

<h2>Leave a Comment</h2>

<form method="post">
    Username:<br />
    <input type="text" name="username" /><br /><br />

    Comment:<br />
    <textarea name="comment" rows="4" cols="50"></textarea><br /><br />

    <input type="submit" value="Submit Comment" />
</form>

<hr />
<h3>All Comments:</h3>

<b>attacker</b>: <script>alert(1)</script><br /><br /><b>attacker</b>: <script>alert(1)</script><br /><br /><b
)</script><br /><br /><b>attacker</b>: <script>alert('Network Security Module')</script><br /><br /><b>admin</
><br /><b></b>: <img src=x onerror=alert(1)><br /><br /><b></b>: <script>alert(1)</script><br /><br /><b></b>:
```

10.7 Attack 7 — VPN Access Validation

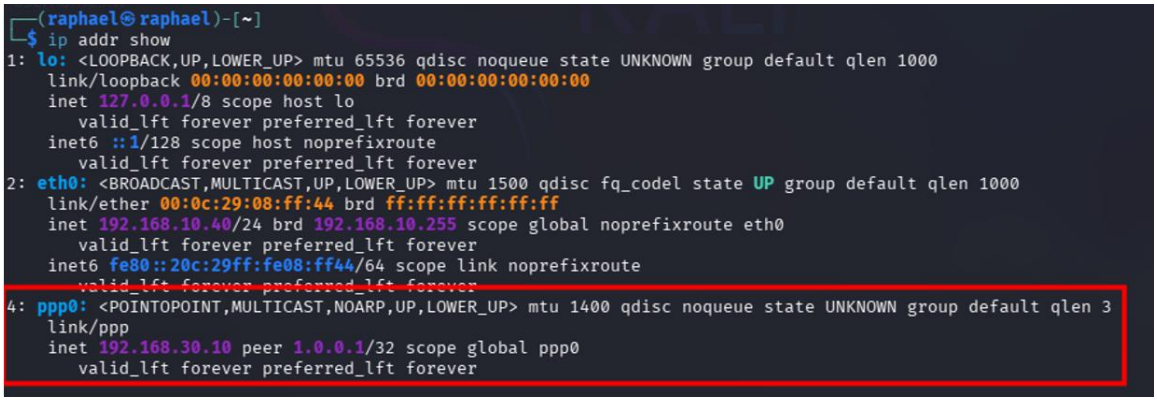
```
nmap -sU -p 500,1701,4500 192.168.10.50    # Confirm VPN ports open
```

Valid credential test: Connect CORP-VPN-Test > ip addr show > look for ppp0 with 192.168.30.x IP > ping 192.168.10.10.

Invalid credential test: Change VPN password to wrong value > attempt connection > should fail. SoftEther Server Manager on VPN01 shows failed auth in server log.

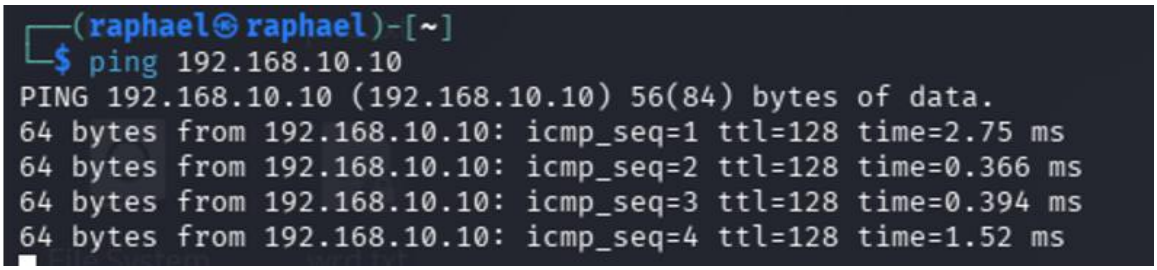


- VPN connection successful



```
┌──(raphael㉿raphael)-[~]
└─$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:08:ff:44 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.40/24 brd 192.168.10.255 scope global noprefixroute eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe08:ff44/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
4: ppp0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1400 qdisc noqueue state UNKNOWN group default qlen 3
    link/ppp
    inet 192.168.30.10 peer 1.0.0.1/32 scope global ppp0
       valid_lft forever preferred_lft forever
```

- Checking if the connection is working or not



```
┌──(raphael㉿raphael)-[~]
└─$ ping 192.168.10.10
PING 192.168.10.10 (192.168.10.10) 56(84) bytes of data.
64 bytes from 192.168.10.10: icmp_seq=1 ttl=128 time=2.75 ms
64 bytes from 192.168.10.10: icmp_seq=2 ttl=128 time=0.366 ms
64 bytes from 192.168.10.10: icmp_seq=3 ttl=128 time=0.394 ms
64 bytes from 192.168.10.10: icmp_seq=4 ttl=128 time=1.52 ms
```

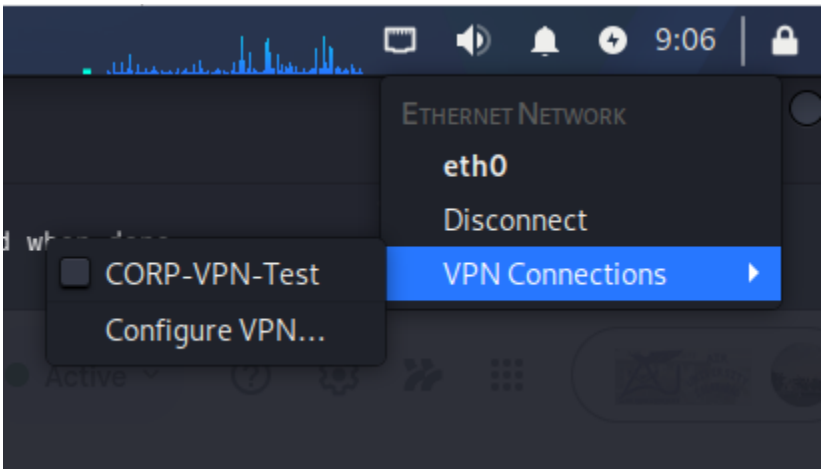
- IDS alerts of SoftEther VPN

```
P} 192.168.10.50:57350 → 192.168.10.10:53
4/23/2026-21:00:16.281606  [**] [1:2068199:1] ET INFO SoftEther VPN Service Related Domain (softether .org) in DNS Lookup [**] [Classification: Misc acti
P} 192.168.10.50:54585 → 192.168.10.10:53
4/23/2026-21:00:17.076898  [**] [1:2068195:1] ET INFO External IP Lookup Domain (get-my-ip-v6 .ddns .softether-network .net) in DNS Lookup [**] [Classifi
[Priority: 3] {UDP} 192.168.10.50:49265 → 192.168.10.10:53
4/23/2026-21:00:17.076898  [**] [1:2068200:1] ET INFO SoftEther VPN Service Related Domain (softether-network .net) in DNS Lookup [**] [Classification: M
: 3] {UDP} 192.168.10.50:49265 → 192.168.10.10:53
4/23/2026-21:00:17.169643  [**] [1:2068195:1] ET INFO External IP Lookup Domain (get-my-ip-v6 .ddns .softether-network .net) in DNS Lookup [**] [Classifi
[Priority: 3] {UDP} 192.168.10.50:49265 → 192.168.10.10:53
4/23/2026-21:00:17.169643  [**] [1:2068200:1] ET INFO SoftEther VPN Service Related Domain (softether-network .net) in DNS Lookup [**] [Classification: M
: 3] {UDP} 192.168.10.50:49265 → 192.168.10.10:53
4/23/2026-21:00:17.410433  [**] [1:2068195:1] ET INFO External IP Lookup Domain (get-my-ip-v6 .ddns .softether-network .net) in DNS Lookup [**] [Classifi
[Priority: 3] {UDP} 192.168.10.50:61854 → 192.168.10.10:53
4/23/2026-21:00:17.410433  [**] [1:2068200:1] ET INFO SoftEther VPN Service Related Domain (softether-network .net) in DNS Lookup [**] [Classification: M
: 3] {UDP} 192.168.10.50:61854 → 192.168.10.10:53
4/23/2026-21:00:21.186038  [**] [1:2068199:1] ET INFO SoftEther VPN Service Related Domain (softether .org) in DNS Lookup [**] [Classification: Misc acti
```

- Nmap services and port scanning for creating noise in the network for IDS detection testing

```
┌──(raphael㉿raphael)-[~]
└─$ nmap -sU -p 500,1701,4500 192.168.10.50
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-02 04:52 -0400
Nmap scan report for 192.168.10.50
Host is up (0.0011s latency).

PORT     STATE         SERVICE
500/udp  open          isakmp
1701/udp closed        L2TP
4500/udp open|filtered nat-t-ike
MAC Address: 00:0C:29:0E:D4:86 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 3.65 seconds
```

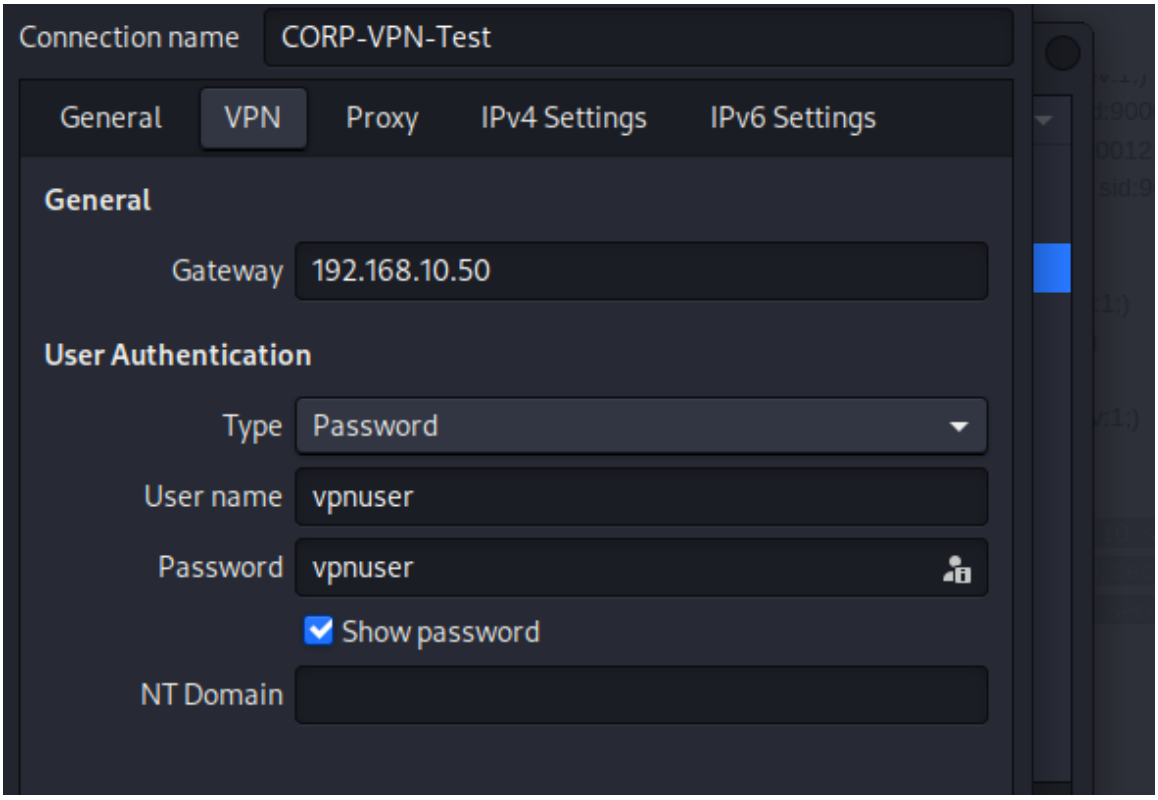
- Connecting with the VPN profile in the Kali linux



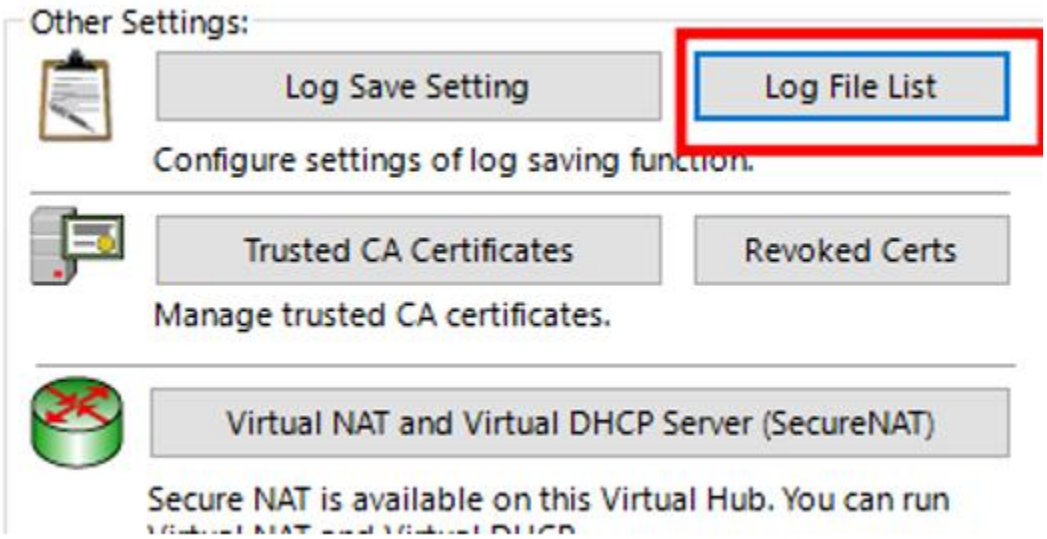
- VPN connection successful

```
└─$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defa
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP gr
    link/ether 00:0c:29:08:ff:44 brd ff:ff:ff:ff:ff:ff
    inet 192.168.10.40/24 brd 192.168.10.255 scope global noprefixroute eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::20c:29ff:fe08:ff44/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
6: ppp0: <POINTOPOINT,MULTICAST,NOARP,UP,LOWER_UP> mtu 1400 qdisc noqueue stat
    link/ppp
    inet 192.168.30.10 peer 1.0.0.1/32 scope global ppp0
       valid_lft forever preferred_lft forever
```

- Entering username and password



- Checking the VPN Log file



- VPN connection logs

Log File Name	File Size	Updated on	Locat...
packet_log/CORP-VPN/pkt_20260315.log	2.43 KBytes	2026-03-15 06:41:10	VPN01
packet_log/CORP-VPN/pkt_20260423.log	6.41 KBytes	2026-04-23 18:48:50	VPN01
security_log/CORP-VPN/sec_20260313.log	511 Bytes	2026-03-13 17:54:31	VPN01
security_log/CORP-VPN/sec_20260315.log	7.32 KBytes	2026-03-15 06:41:10	VPN01
security_log/CORP-VPN/sec_20260423.log	11.54 KBytes	2026-04-23 18:54:36	VPN01
security_log/DEFAULT/sec_20260313.log	113 Bytes	2026-03-13 17:33:23	VPN01

- Here we can see which user is connected to the VPN and which user is trying to bypass the password security, Connection Failed

```
18:53:41.627 Connection "CID-5-8D510E249E": User authentication failed. The user r
18:54:23.284 The connection "CID-6-9AAA9D80F7" (IP address: 192.168.10.40, Host na
18:54:23.284 Connection "CID-6-9AAA9D80F7": User authentication failed. The user r
18:54:36.362 The connection "CID-7-368C2F96E7" (IP address: 192.168.10.40, Host na
18:54:36.362 Connection "CID-7-368C2F96E7": User authentication failed. The user r
```

10.8 Attack 8 — TLS Version Verification

Goal: Confirm TLS 1.0 and 1.1 are rejected; only TLS 1.2 accepted.

```
nmap --script ssl-enum-ciphers -p 636 192.168.10.10     # LDAPS
nmap --script ssl-enum-ciphers -p 3389 192.168.10.10    # RDP
```

Expected Result: Only TLSv1.2 appears in output. TLSv1.0 and TLSv1.1 not listed.

- Nmap ssl-enum cipher attack

```
┌──(raphael㉿raphael)-[~]
└─$ nmap --script ssl-enum-ciphers -p 3389 192.168.10.10
Starting Nmap 7.98 ( https://nmap.org ) at 2026-05-02 04:56 -0400
Nmap scan report for 192.168.10.10
Host is up (0.00078s latency).

PORT     STATE    SERVICE
3389/tcp filtered ms-wbt-server
MAC Address: 00:0C:29:D0:E8:38 (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1.03 seconds

┌──(raphael㉿raphael)-[~]
```

11. Monitoring and Logging Verification

11.1 Sysmon Verification (DC01)

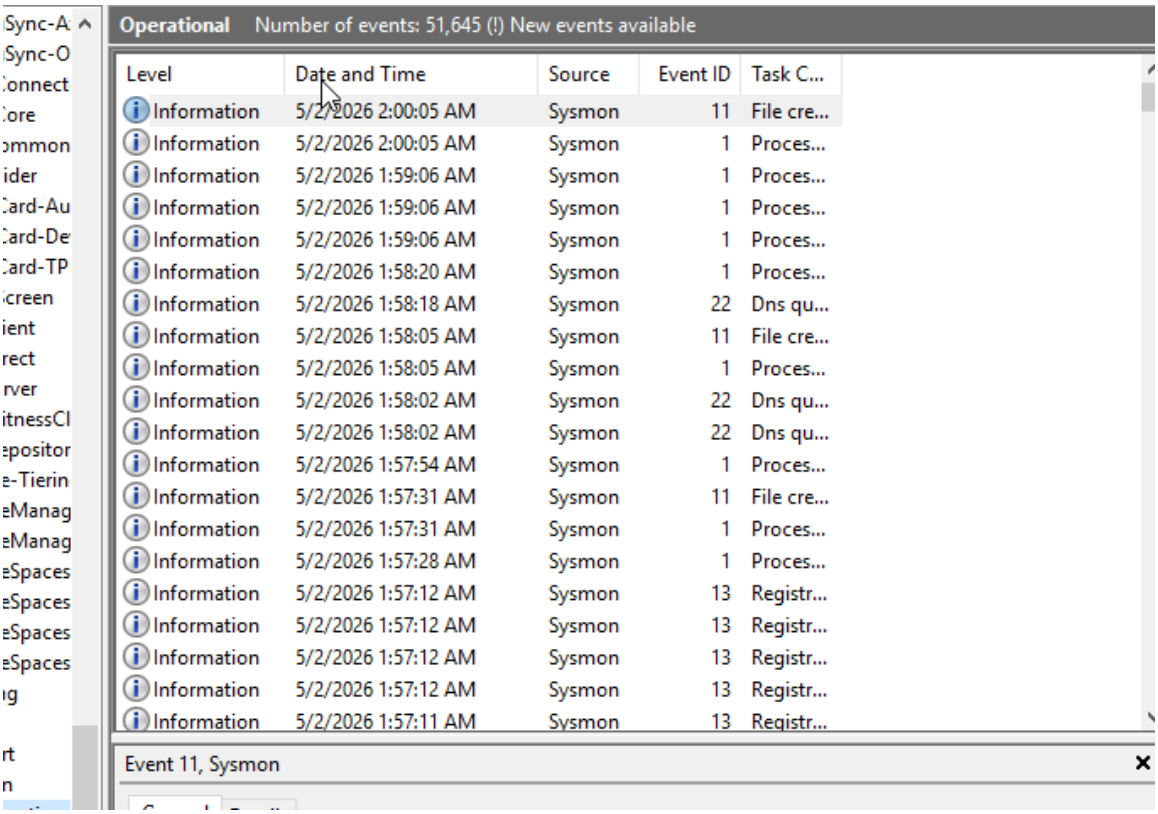
```
Get-Service sysmon64    # Expected: Status = Running
```

Event Viewer path for Sysmon logs: Applications and Services Logs > Microsoft > Windows > Sysmon > Operational

Sysmon was confirmed running on DC01. Security log (128 MB) and System log (6.07 MB) were active with recent entries as of 5/2/2026.

```
PS C:\Users\Administrator> Get-Service sysmon64

Status   Name               DisplayName
------   ----               -----------
Running  Sysmon64           sysmon64
```

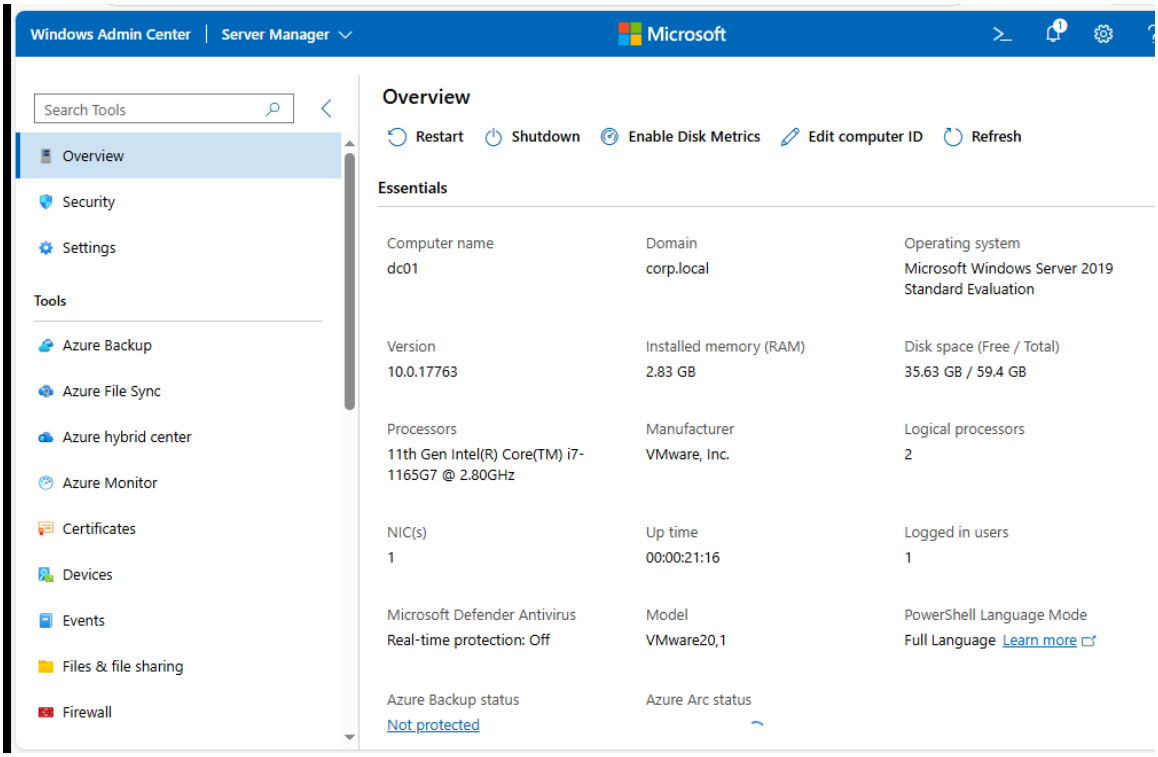
- Sysmon Logs correlation



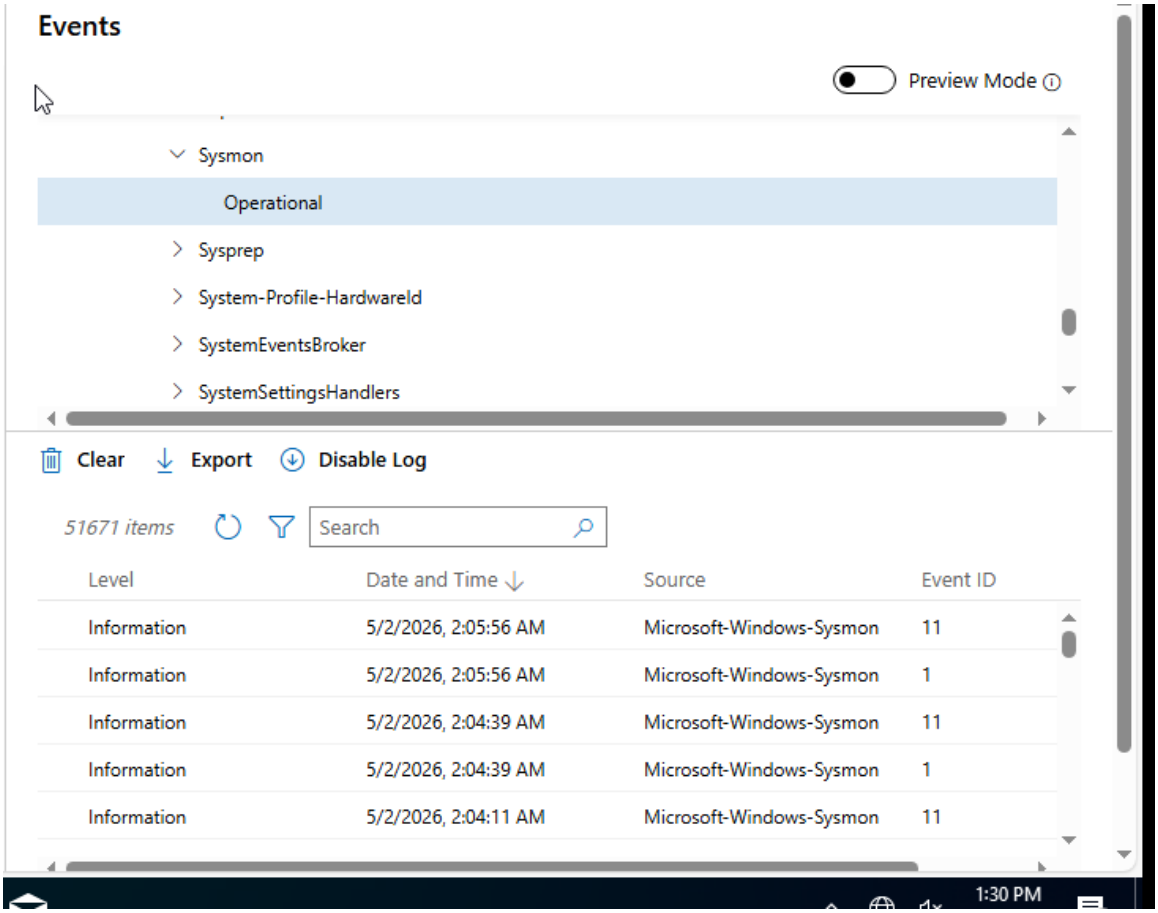
11.2 Windows Admin Center (ADMIN01)

Accessed at https://localhost:6516. Verified:

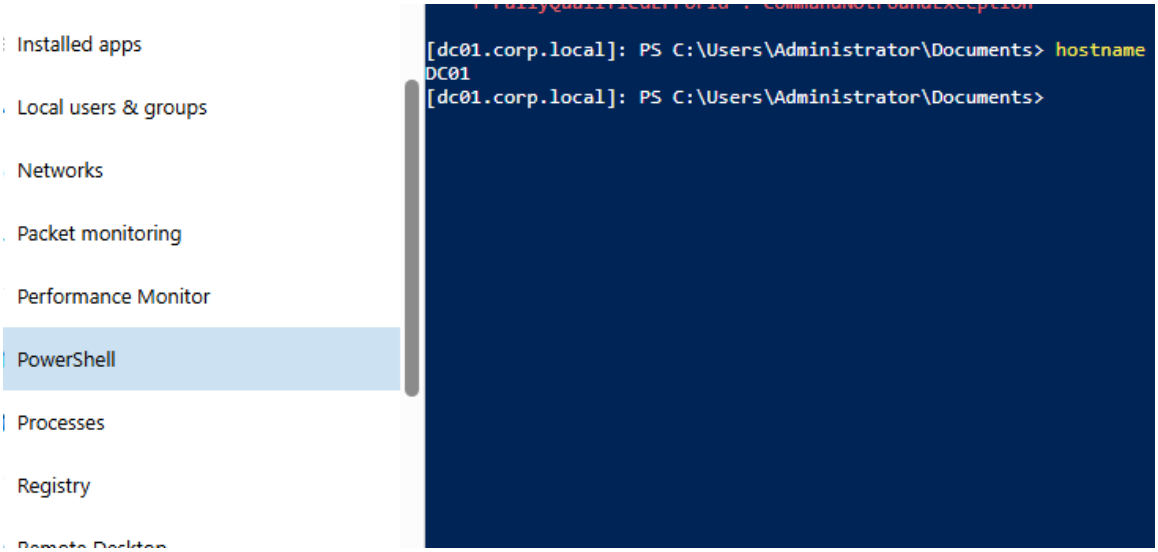
- DC01 visible in server list with CPU and RAM overview
- Events tab showing Security logs loading
- PowerShell tab: hostname command returned DC01



- Go to Sysmon→ Operational

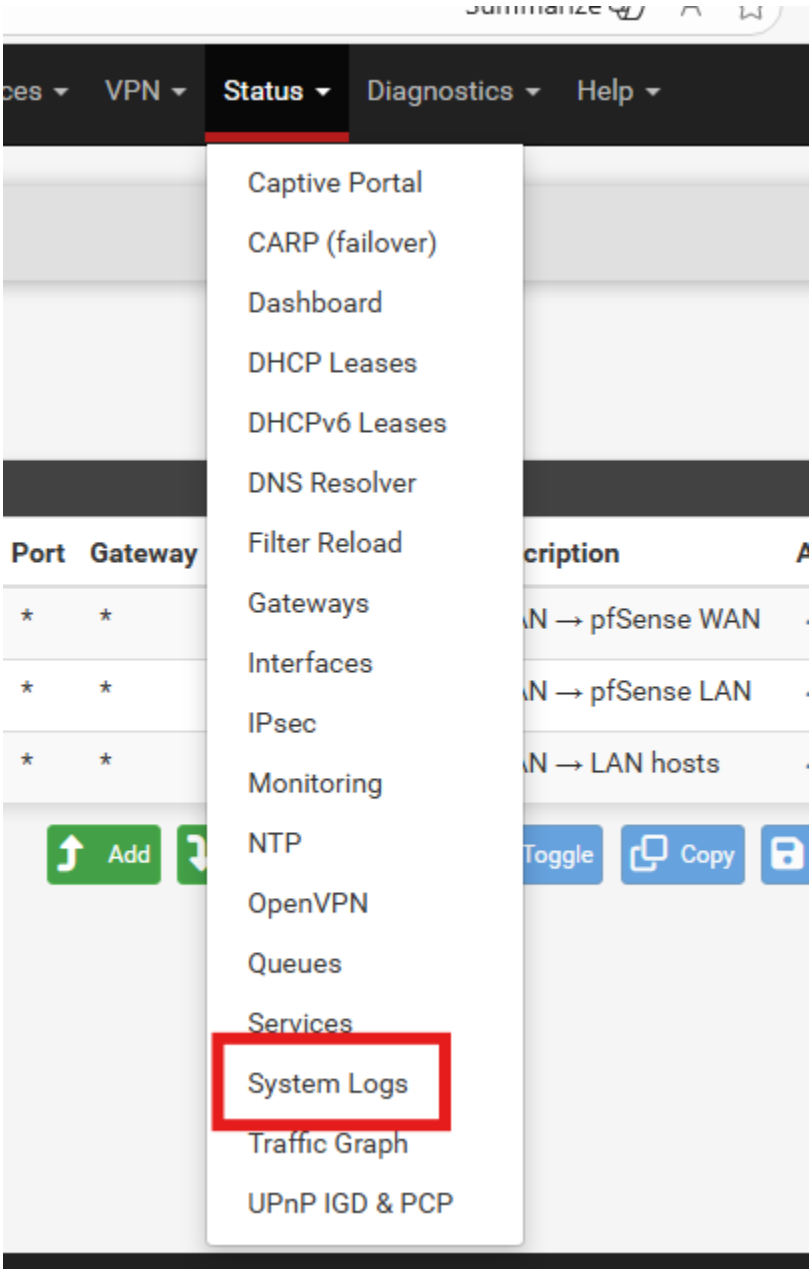


- Open powershell through windows admin center



11.3 pfSense Firewall Logs

Firewall logs monitored at Status > System Logs > Firewall. Dynamic view used (status_logs_filter_dynamic.php) as log volume exceeded standard view limits during attack simulation. Logs confirmed blocked ICMP and RDP traffic from CL01, and allowed AD service traffic.

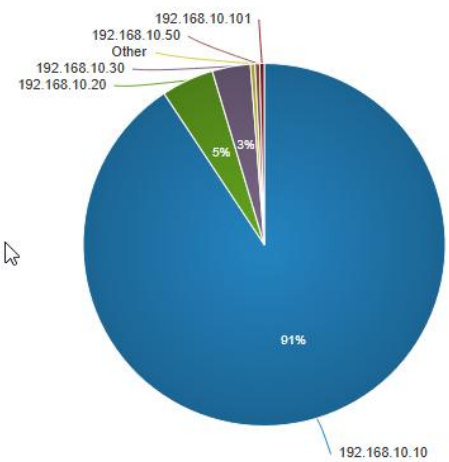


- Pfsense Logs, All the allowed and blocked rules triggered

May 1 15:05:00	kernel		TSC: P-state invariant
May 1 15:05:00	kernel		Hypervisor: Origin = "VMwareVMware"
May 1 15:05:00	kernel		intsmb0: <Intel PIIX4 SMBUS Interface> port 0x1040-0x104f at device 7.3 on pci0
May 1 15:05:00	kernel		intsmb0: intr SMI disabled revision 0
May 1 15:05:00	kernel		smbus0: <System Management Bus> on intsmb0
May 1 15:05:00	kernel		lo0: link state changed to UP
May 1 15:05:00	sshguard	16768	Now monitoring attacks.
May 1 15:05:02	kernel		done.
May 1 15:05:02	php-fpm	435	/rc.linkup: The command '/sbin/dhclient -c /var/etc/dhclient_wan.conf -p /var/run/dhcli em0 > /tmp/em0_output 2> /tmp/em0_error_output' returned exit code '15', the output w
May 1 15:05:02	check_reload_status	506	rc.newwanip starting em0
May 1 15:05:03	php-fpm	436	/rc.newwanip: rc.newwanip: Info: starting on em0.
May 1 15:05:03	php-fpm	436	/rc.newwanip: rc.newwanip: on (IP address: 192.168.1.128) (interface: WAN[wan]) (real m0).
May 1 15:05:03	kernel		pflog0: promiscuous mode enabled
May 1 15:05:04	php-fpm	436	/rc.newwanip: Gateway, NONE AVAILABLE
May 1 15:05:04	php-fpm	436	/rc.newwanip: Gateway, none 'available' for inet6, use the first one configured. 'WAN_DH

- Source Ips chart

Source IPs



Source IPs	Data points	
192.168.10.10	9062	Lookup
192.168.10.20	469	Lookup
192.168.10.30	341	Lookup
192.168.10.50	41	Lookup
192.168.10.101	40	Lookup

11.4 Suricata IDS Logs (IDS01)

Live monitoring via:

```
sudo tail -f /var/log/suricata/fast.log
```

During attack simulation, the following alert categories were detected and logged:

- Brute Force — SMB authentication attempts from ATTACK01
- TCP Port Scan — nmap reconnaissance detected
- ET Scan — Emerging Threats scan signatures triggered
- NBT-NS Query — NetBIOS activity detected on the network

```
┌──(cyber㉿cyber)-[/var/lib/suricata/rules]
└─$ sudo tail -f /var/log/suricata/fast.log
[sudo] password for cyber:
04/23/2026-21:02:16.186577  [**] [1:2068199:1] ET INFO So
DP} 192.168.10.50:62777 → 192.168.10.10:53
```

- Checking for IDS logs
- Brute force Attack detected

```
05/01/2026-16:03:59.731438  [**] [1:9000032:1] Brute Force - SMB Authentication [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:38798 → 192.168.10.10:445
05/01/2026-16:03:59.743571  [**] [1:2260000:1] SURICATA Applayer Mismatch protocol both directions [**] [Classification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.10.40:3836
68.10.10:135
05/01/2026-16:03:59.747575  [**] [1:9000003:1] SMB Brute Force Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.
05/01/2026-16:03:59.747575  [**] [1:9000032:1] Brute Force - SMB Authentication [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:38860 → 192.168.10.10:445
05/01/2026-16:03:59.758403  [**] [1:9000003:1] SMB Brute Force Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.
05/01/2026-16:03:59.758403  [**] [1:9000032:1] Brute Force - SMB Authentication [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:38938 → 192.168.10.10:445
05/01/2026-16:03:59.764700  [**] [1:9000003:1] SMB Brute Force Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:38992 → 192.168.10.10:445
05/01/2026-16:03:59.764700  [**] [1:9000032:1] Brute Force - SMB Authentication [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:38992 → 192.168.10.10:445
05/01/2026-16:03:59.773437  [**] [1:9000003:1] SMB Brute Force Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:39040 → 192.168.10.10:445
05/01/2026-16:03:59.773437  [**] [1:9000032:1] Brute Force - SMB Authentication [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:39040 → 192.168.10.10:445
05/01/2026-16:03:59.773439  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:58466 → 192.168.10.10:464
05/01/2026-16:03:59.779037  [**] [1:9000003:1] SMB Brute Force Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:39068 → 192.168.10.10:445
05/01/2026-16:03:59.779037  [**] [1:9000032:1] Brute Force - SMB Authentication [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:39068 → 192.168.10.10:445
05/01/2026-16:04:04.279557  [**] [1:2068199:1] ET INFO SoftEther VPN Service Related Domain (softether .org) in DNS Lookup [**] [Classification: Misc activity] [Priority: 3] {UDP} 192.168.10.5
 192.168.10.10:53
```

- Nmap Scan logs

```
 192.168.10.10:53
05/01/2026-16:07:36.113612  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:9666
05/01/2026-16:07:36.413670  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:9593
05/01/2026-16:07:36.523463  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:49155
05/01/2026-16:07:36.717649  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:1038
05/01/2026-16:07:36.731264  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:44443
05/01/2026-16:07:36.829550  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:5405
05/01/2026-16:07:36.929820  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:541
05/01/2026-16:07:37.017692  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:5080
05/01/2026-16:07:37.041827  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:6543
05/01/2026-16:07:37.135504  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:6003
05/01/2026-16:07:37.237698  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:1057
05/01/2026-16:07:37.311771  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:5631
05/01/2026-16:07:37.347714  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:7070
05/01/2026-16:07:37.437570  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:6123
05/01/2026-16:07:37.452291  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:2382
05/01/2026-16:07:37.544475  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:6059
05/01/2026-16:07:37.639332  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:40911
05/01/2026-16:07:37.657762  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60834 → 192.168.10.10:1091
05/01/2026-16:07:37.754927  [**] [1:9000002:1] TCP Port Scan Detected [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.10.40:60832 → 192.168.10.10:9110
```

- SQL Injection detected

```
05/01/2026-16:01:59.872452  [**] [1:9000042:1] NBT-NS Query Detected - Possible Poisoning [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.10.40:36918 → 192.168.10.10:137
05/01/2026-16:02:01.154210  [**] [1:9000042:1] NBT-NS Query Detected - Possible Poisoning [**] [Classification: (null)] [Priority: 3] {UDP} 192.168.10.40:49900 → 192.168.10.10:137
05/01/2026-16:02:04.192680  [**] [1:2068199:1] ET INFO SoftEther VPN Service Related Domain (softether .org) in DNS Lookup [**] [Classification: Misc activity] [Priority: 3] {UDP} 1
 192.168.10.10:53
05/01/2026-16:02:09.143741  [**] [1:2068199:1] ET INFO SoftEther VPN Service Related Domain (softether .org) in DNS Lookup [**] [Classification: Misc activity] [Priority: 3] {UDP} 1
```

12. Issues Encountered and Resolutions

Issue	Root Cause	Resolution
IDS01 DNS resolution failing	/etc/resolv.conf empty or had extra space	echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
Suricata upgrade stuck for 2+ hours	No internet access / DNS broken on IDS01	Cancelled upgrade (Ctrl+C), skipped — not required for module
ATTACK01 VPN ppp0 DOWN	xl2tpd service not installed/running	Installed network-manager-l2tp packages, started xl2tpd service
local.rules empty	Default Suricata install has empty local.rules — rules must be added manually	Manually created and populated local.rules with all 15 custom rules
nmcli error: unknown connection CORP-VPN-TEST	Command used wrong case (all caps) vs saved name (CORP-VPN-Test)	Used correct case: nmcli con up "CORP-VPN-Test"
pfSense LAN rules not working	Old rules from previous modules still present	Deleted old rules, recreated clean rule set per Module 4 requirements

13. Summary and Conclusion

Security Control	Component	Status
Edge Firewall	pfSense — LAN rules enforced	IMPLEMENTED
Block ICMP from CL01	pfSense Rule 1	VERIFIED
Block RDP from CL01 to DC01	pfSense Rule 2	VERIFIED
Allow AD Services	pfSense Rules (8 ports)	VERIFIED
Allow VPN Client Traffic	pfSense VPN Rule	IMPLEMENTED
SMBv1 Disabled	Module 3 GPO — still active	VERIFIED
Account Lockout Policy	Module 3 GPO — still active	VERIFIED
TLS 1.0 Disabled	Module 3 Registry — still active	VERIFIED
LLMNR Disabled	Module 3 GPO — still active	VERIFIED
SQL Injection Detection	Suricata IDS Rules	IMPLEMENTED
XSS Detection	Suricata IDS Rules	IMPLEMENTED
Brute Force Detection	Suricata IDS Rules	IMPLEMENTED
LLMNR Poisoning Detection	Suricata IDS Rules	IMPLEMENTED
Sysmon Logging	DC01 — active and logging	VERIFIED
VPN Server	SoftEther — CORP-VPN Online	VERIFIED

> **Module 4 successfully adds the perimeter security layer that completes the three-module lab series.**

With pfSense as the central gateway, traffic policies are enforced at the network level rather than relying solely on host-based controls. The attack simulation confirmed that each security layer works as intended:

- The firewall blocks restricted traffic (ICMP and RDP from CL01)
- The IDS detects application-layer attacks (SQL injection, XSS, brute force, LLMNR)
- The account lockout policy stops brute force credential attacks
- LLMNR poisoning fails due to the GPO applied in Module 3
- SMBv1 / EternalBlue attack fails due to Module 3 hardening
- TLS 1.0 and 1.1 are rejected — only TLS 1.2 is accepted

Together, Modules 2, 3, and 4 demonstrate a practical and functional implementation of defence in depth across administrative, technical, and perimeter security controls in a fully virtualised lab environment.